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Early-Warning Industrial Fault Detection based on Physics-Guided Residual Learning and Calibrated CRNNs

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ABSTRACT

Industrial plants generate high volume multivariate sensor time series where early fault detection must be accurate, governable and explainable under regime changes, sensor correlations and drift. Many published Tennessee Eastman Process (TEP) studies report strong receiver operating characteristic (ROC) or accuracy but provide limited evidence for calibrated probabilities, uncertainty bounds and actionable sensor level diagnostics, so alarm thresholds remain ad hoc and difficult to defend in control rooms. To address this issue, we propose a physics guided pipeline that combines twin based residuals with compact rolling and spectral descriptors and an attention based convolutional recurrent neural network (CRNN). The proposed method is also compared against state-of-the-art (SOTA) baselines to establish its relative effectiveness under the same evaluation setting. Residuals highlight deviations from physics consistent behavior, while a one-dimensional convolutional neural network (Conv1D) and bidirectional gated recurrent unit (BiGRU) encoder with attention models short and long temporal structure over 120 step windows. SHapley Additive exPlanations (SHAP) guided feature selection reduces redundancy and improves interpretability, Optuna stabilizes training and Platt scaling calibrates anomaly probabilities for policy driven thresholding under a false alarm budget. Reliability is quantified with 1000 block bootstrap bias-corrected and accelerated (BCa) confidence intervals. On a run level 70/20/10 split of Tennessee Eastman Process runs, the method achieves about 99.0% accuracy with area under the receiver operating characteristic curve (AUC-ROC) and area under the precision-recall curve (AUC-PR) near
1.00, macro F1-score of 0.93 ± 0.02 and expected calibration error (ECE) reduced to ≈ 0.03 after calibration. Early warning
governance improves the Numenta Anomaly Benchmark (NAB) score by 17% over an Isolation Forest baseline and supports
low nuisance alarm operation. These results show that residual-guided learning plus calibrated decision governance can support deployment-oriented industrial monitoring by enabling auditable thresholds and operator-facing explanations within the Tennessee Eastman benchmark setting.

KEYWORDS: Industrial fault diagnosis, Physics-guided residuals, Convolutional recurrent neural network (CRNN), Probabil-ity calibration

• Introduction

Modern manufacturing environments utilize vision systems, vibration sensors, power monitors and Internet of Things (IoT)-enabled devices to generate extensive, rapidly evolving multivariate time-series data streams.[1] Early detection of anomalies within these streams is essential to prevent product defects, unplanned downtime and safety-critical incidents, which are central concerns in achieving zero-defect Industry 4.0 operations.[2] The economic implications are significant.[3] Siemens

estimates that the 500 largest global companies lose nearly United States dollars (USD) 1.4 trillion annually due to unplanned downtime, representing approximately 11% of their revenues and directly impacting competitiveness and risk exposure[4]. At the operational level, reliability surveys indicate that unplanned outages are both frequent and costly, with typical losses approaching USD 125,000 per hour and most industrial businesses experiencing outages at least once per month[5]. This urgency is further underscored by the rapid expansion of the predictive maintenance market, which is projected to increase from USD 10.6 billion in 2024 to USD 47.8 billion by 2029, demonstrating strong demand for reliable and actionable monitoring systems[6].

• Industrial anomaly detection under deployment constraints

Although significant progress has been made in statistical process control, machine learning and deep learning, a substantial gap persists between research prototypes and systems suitable for deployment and defense in production environments.[7] Industrial plants frequently experience changes in operating modes, sensor drift, correlated measurements and evolving process dynamics, while labeled fault data remain scarce and costly to obtain.[8] Fixed rules and static thresholds often prove inadequate under these conditions, whereas purely data-driven deep learning models may lack transparency, stable alarm rates and audit-ready evidence required by operators.[9] Additional deployment constraints further restrict the design space: alerts must be generated with low latency, memory usage must accommodate edge hardware limitations and outputs must be interpretable and actionable to enable engineering teams to localize issues to specific sensors, time intervals and plausible physical causes.[10]

Model-based simulation offers a complementary source of evidence by generating expected behavior under nominal conditions.[11] Comparing real-time measurements to these predictions produces residual signals that highlight deviations from expected dynamics and can reveal subtle faults earlier than raw signals alone.[12] Despite these advantages, many residual-based methods continue to rely on ad hoc thresholding and do not systematically integrate residuals with compact descriptors, advanced temporal learning models, calibrated probability outputs or reproducible evaluation frameworks that link technical metrics to operational decisions.[13] These limitations underscore the need for hybrid pipelines that are both statistically rigorous and operationally practical.[14] In such pipelines, residual-informed inputs enhance sensitivity, calibrated probabilities enable defensible threshold selection and interpretable explanations facilitate investigation workflows.[15]

Figure 1 summarizes a typical industrial workflow, from multichannel data acquisition to automated defect decisions and downstream quality-control actions.

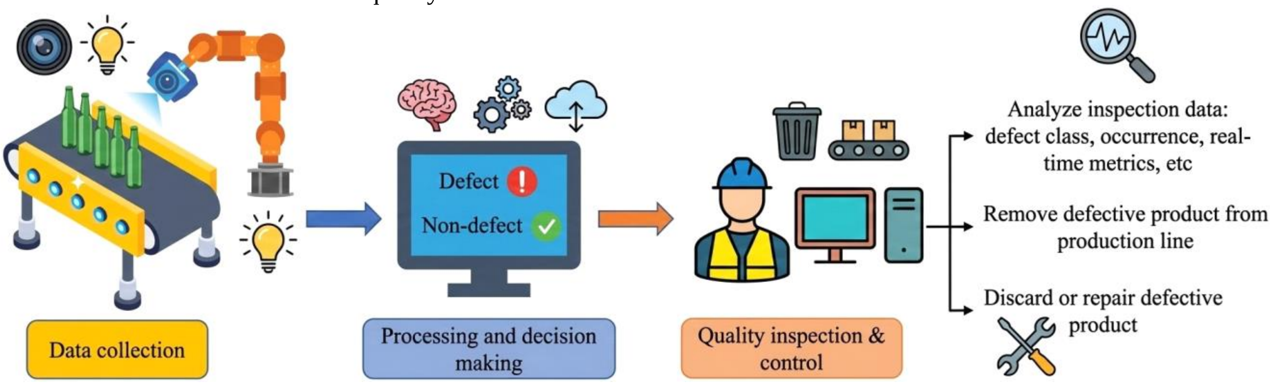


Figure 1. End-to-end industrial anomaly detection and quality inspection workflow. This figure is an enhanced version adapted from[16], illustrating data collection from production lines, automated defect and non-defect decision making, and downstream quality inspection and control actions.

• Residual-guided learning with calibrated decisions

A unified anomaly detection pipeline is introduced and evaluated using the Tennessee Eastman Process dataset, a widely recognized benchmark for process-industry fault diagnosis.[17] The pipeline employs a physics-based model to generate physics-informed expectations and computes residuals as the difference between observed measurements and model predictions.[18] These residuals provide an interpretable feature signal that enhances sensitivity to subtle faults and eliminates the need to develop a new simulator for each study.[19] The residuals are integrated with compact engineered descriptors and processed by a lightweight convolutional recurrent architecture with attention, which is designed to capture both short-term temporal structure and longer-range dependencies while remaining suitable for edge deployment.[20]

Decision quality is established as a fundamental requirement rather than an afterthought.[21] Predicted probabilities are calibrated and operating points are determined through threshold-sensitivity analyses that report achievable true positive rates, false positive rates and alarm rates per hour, thereby linking technical performance to control-room decision-making.[22]

Reliability is assessed using expected calibration error and bootstrap confidence intervals, ensuring that uncertainty and threshold decisions can be justified to operators and auditors rather than adjusted through trial and error.[23] To facilitate investigation workflows, the pipeline provides per-fault performance metrics and stable feature attributions, enabling alarms to be associated with specific sensors and time spans for root-cause analysis and auditable maintenance actions.[24]

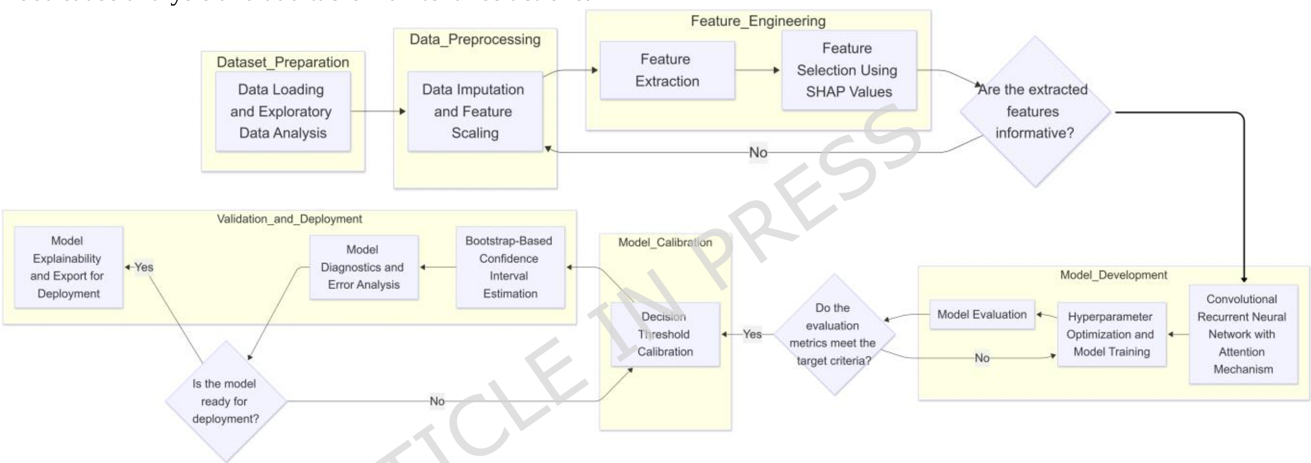


Figure 2. End-to-end methodology flowchart showing physics-model residual generation, residual-enriched input construction, attention-based convolutional recurrent training, probability calibration and validation with per-fault reporting and attribution stability checks.

• Scope and paper organization

Focusing on the Tennessee Eastman Process improves comparability with prior work and enables controlled analysis of regime changes and fault families. At the same time, simulation-based evidence has limits, so the results are framed as a strong internal-validity step toward field deployment rather than a definitive proof of operational performance. The remainder of this study is organised as follows: Section 2 reviews related work, Section 3 details the proposed methodology and Section 4 reports experimental results. Section 5 discusses findings and comparisons, while Section 6 concludes with limitations and future directions. Tables 16 and 17 summarize the variables, symbols, acronyms and abbreviations used in this manuscript.

• Literature Review

This section reviews anomaly detection for smart manufacturing with emphasis on the core deployment problem highlighted in the Introduction, namely regime changes, drift, limited labels, low-latency requirements and the need for calibrated and explainable alarms.

• Classical monitoring before deep learning: statistical tests and residual-based diagnostics

Early industrial monitoring relied on statistical process control and multivariate statistical process monitoring, where control limits were defined for summary statistics under approximate stationarity.[25] In the Tennessee Eastman Process, this line of work includes principal component analysis (PCA), dynamic

PCA, partial least squares (PLS)-style monitoring and knowledge-guided classifiers that translate multivariate deviations into alarm decisions.[26] These methods remain attractive because they are lightweight and often easier to communicate to operators.[27] However, their assumptions are routinely violated in real plants, where operating modes shift, sensors drift, interactions are nonlinear and fault signatures evolve over time.[28] As a result, fixed thresholds can become unstable, producing either missed detections under new regimes or elevated false alarms under benign distribution shifts.[26] Residual-based monitoring partially addresses this issue by comparing measured behavior to expected behavior and alarming on deviations, but classical residual tests still depend heavily on hand-tuned limits and may degrade when residual distributions change across operating conditions.[29]

- **Residual-guided deep learning for regime changes and edge feasibility**

Recent research has increasingly focused on temporal deep learning to more effectively capture the nonlinearity and long-range dependencies present in multivariate industrial data streams.[30] Convolutional temporal models are capable of identifying local patterns, such as transient spikes and short-lived oscillations, whereas recurrent layers enhance sensitivity to delayed effects and gradual drifts.[31,32] Attention mechanisms further address regime variability by directing computational resources

toward informative time segments instead of uniformly processing all timestamps.[33] Concurrently, unsupervised and weakly supervised deep anomaly detection methods have been extensively investigated using Tennessee Eastman style datasets to mitigate reliance on costly fault labels.[34] A comprehensive benchmark on Tennessee Eastman Process data demonstrates that several representative deep anomaly detectors achieve best F1-scores of 0.9172, 0.9114, 0.9097, 0.9078 and 0.9074 for temporal forecasting and hybrid graph or recurrent approaches, indicating that strong performance is attainable but not consistent across different methodological families.[14] These results are practically significant because they represent realistic trade-offs among model classes and training procedures, rather than the performance of a single, hand-optimized architecture.[35]

While pure deep learners can be accurate, two gaps remain relative to the deployment problem emphasized in the Introduction.[36,37] First, many approaches treat the input space as purely data-driven and do not explicitly encode physics-informed expectations that can make subtle deviations more visible and more interpretable.[38] Second, many studies prioritize discrimination metrics while providing limited support for decision governance, such as calibrated probabilities, threshold justification and operator-facing reporting.[39] Residual-guided learning addresses the first gap by using a physics-based predictor to define expected behavior and then learning on residual-enriched representations, which improves sensitivity to small drifts that may be masked in raw signals.[40] Compact hybrid architectures address the second gap by balancing expressiveness with edge feasibility, using convolutional blocks for short-range structure, gated recurrent units (GRUs) for longer dependencies and attention to emphasize informative windows without excessive compute.[41] Feature selection further improves deployability and interpretability by reducing redundancy and highlighting sensor-level relevance, which supports maintenance workflows that must map alarms to likely causes and affected subsystems.

As shown in Figure 3, the red box highlights the reactor-condenser recycle loop, a tightly coupled subsystem where nonlinear interactions, short transients and delayed responses can produce weak but safety-critical deviations. Our framework targets this region by forming physics-informed residual signals that amplify departures from expected behavior, then modeling the residual-enriched multivariate streams with a compact convolutional-recurrent backbone and attention to capture both short-lived oscillations and long-range drift. In addition, calibrated probability outputs and threshold governance provide stable, operator-defensible alarms, while feature selection supports sensor-level interpretability for maintenance handoff.

- **Trustworthy alarms: calibration, uncertainty and explanation for operational handoff**

Beyond detection accuracy, industrial anomaly systems must deliver probability outputs that are reliable enough to support defendable thresholding and stable alarm policies.[43] Calibration has therefore become an enabling layer for governance, since miscalibrated confidence can lead to overconfident alarms and unstable decision boundaries under shift.[38] Post hoc calibration methods, including Platt-style scaling and related temperature variants, are widely used to align predicted confidence with empirical correctness and calibration quality is commonly summarized via reliability curves and expected calibration error (ECE).[44] For time series in particular, uncertainty reporting should respect temporal dependence, so interval estimation via block-aware bootstrap resampling is often preferred over single split reporting when the goal is to justify operating points.[45] Interpretability is similarly essential for adoption in control-room settings, where engineers require evidence that an alarm is linked to specific sensors and time spans.[46,47] Attribution methods such as SHapley Additive exPlanations (SHAP)-style feature relevance and gradient-based temporal attributions can support this need, but their use in industrial monitoring is most credible when stability across random seeds and training variability is reported.[48] Finally, deployment-oriented evaluation increasingly includes hardware telemetry such as latency, throughput, parameter count and memory footprint, because edge feasibility is not implied by accuracy alone.[49] Together, calibration, uncertainty reporting and stable explanations form the practical interface between a detection model and the investigation workflows required for safe industrial operation.[40] Residual generation based on the mismatch between measured plant outputs and model-based nominal predictions is a classical principle in fault detection and isolation (FDI) rather than a new methodological contribution of this work. In the analytical-redundancy and model-based FDI literature, residuals have long been used as the primary signal for detecting departures from expected system behavior[50–54]. Accordingly, the novelty of the present study does not lie in introducing residual-based detection itself, but in integrating a classical residual-generation stage with calibrated convolutional recurrent neural network (CRNN)-based temporal modeling, SHAP-guided feature selection and governance-oriented threshold design for early warning in industrial monitoring.

• Methodology

This section presents the proposed physics-guided convolutional recurrent neural network (CRNN) pipeline for industrial anomaly detection. The design addresses the deployment constraints introduced in Section 1, including regime changes, drift, limited labels, low latency and operator-facing interpretability. The complete workflow shown in Figure 2 is implemented as a modular sequence comprising residual generation, feature construction, feature selection, temporal learning, probability calibration and validation.

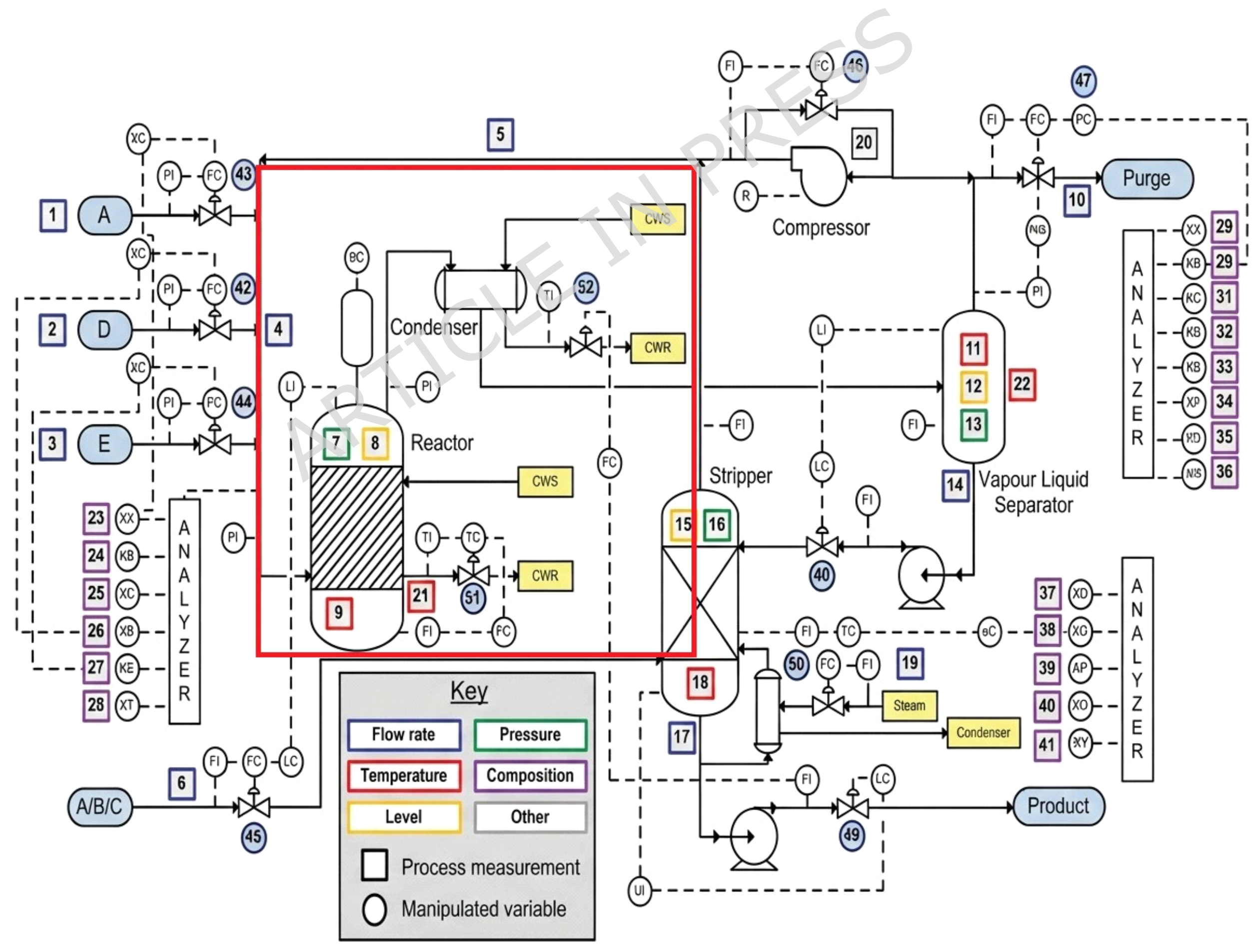


Figure 3. Tennessee Eastman Process schematic with the highlighted region of interest. The red box marks the reactor-condenser recycle loop and its associated feed and utility interactions, where coupled dynamics, operating-mode changes and sensor correlations often cause subtle drifts and delayed fault effects that are

difficult to govern with fixed thresholds. The schematic is adapted and enhanced from the original Tennessee Eastman Process diagram reported in[42].

- **Pipeline overview**

The pipeline starts from multivariate sensor streams of the Tennessee Eastman Process and a physics-based reference model that provides nominal fault-free expectations. Residuals are computed from the mismatch between measured and expected behavior, then combined with compact time- and frequency-domain descriptors. A Light Gradient Boosting Machine (LightGBM) stage is used for SHAP-guided feature selection to reduce redundancy and improve interpretability. The selected features are subsequently processed by the proposed physics-guided CRNN with attention, which outputs calibrated anomaly probabilities for threshold-based decision support. Figure 5 summarizes the data path up to model entry and Figure 6 summarizes the discrimination and calibration evaluation pipeline.

- **Dataset and split**

We evaluate the proposed framework on the Tennessee Eastman Process dataset, which is a widely used benchmark for process-industry fault diagnosis[55,56]. It contains 41 measured variables and 12 manipulated variables across 22 operating conditions, including one normal mode and 21 fault modes. Table 1 summarizes the dataset characteristics used throughout this study. A run-level 70/20/10 split is used for training, validation and testing to avoid temporal leakage across runs. All normalization statistics and feature-selection models are fitted on the training split only and then applied unchanged to validation and test partitions. The learning problem is formulated as binary fault detection rather than explicit fault-type diagnosis. Accordingly, the model outputs a calibrated probability of abnormal operation, whereas fault identities are used only for stratified analysis and error breakdowns. Since the Tennessee Eastman Process is a simulated benchmark rather than a real plant dataset, abnormal

conditions are introduced through benchmark-defined IDV disturbances rather than a single generic corruption mechanism such as additive Gaussian noise or a fixed percentage bias.

Table 1. Summary of key characteristics of the Tennessee Eastman Process dataset used in this study

Property	Type	Description or value	Notes
Total samples	Count	281,170	Combined across 500 runs
Number of simulation runs	Count	500	Distinct operating conditions
Time series length per run	Range	500-1000 steps	Varies with run and fault duration
Process variables xmeas	Sensor inputs	41	Continuous measurements
Manipulated variables xmv	Control inputs	12	Actuator signals
Number of fault types	Integer	21 plus normal	IDV 0 normal, IDV 1-21 faults
Fault label variable	Categorical	`faultNumber` in 0..21	Encodes specific fault type
Class label variable	Binary	`Class` in {FaultFree, Faulty}	Derived from faultNumber
Class distribution	Ratio	~50% FaultFree / 50% Faulty	Overall balance; per-fault imbalance exists
Missing values	Boolean	None observed	No missing entries in provided data
Per-fault imbalance	Qualitative	Present	Frequencies vary across faults

- **Data preprocessing and quality control**

Before residual generation and feature extraction, the multivariate Tennessee Eastman Process sequences are passed through a lightweight preprocessing and quality-control pipeline designed to preserve fault signatures while maintaining numerical stability. Table 2 summarizes the adopted steps. Because the benchmark is simulator-generated, the released sequences were structurally complete after loading and alignment, so missing-data imputation was not required in the final experiments. Nevertheless, the imputation rule is defined explicitly for completeness and reproducibility. Similarly, no aggressive outlier-removal stage was applied because large excursions may correspond to true fault behavior rather than spurious corruption. Instead, training-set-based normalization was used to stabilize optimization without suppressing abnormal dynamics.

Table 2. Preprocessing and quality-control summary applied before residual generation and feature construction.

Stage	Operation	Purpose / policy
	Integrity check	Verify channel count, sequence length, label alignment and time ordering

Ensures that each run is structurally valid before model processing

Missing-data audit imputation rule is	Check all channels for NaN, Inf or undefined entries	Benchmark sequences were complete after loading; fallback defined for reproducibility
Outlier policy onset rather than	No hard sample removal or winsorization	Preserves abrupt deviations that may correspond to true fault noise
Normalization dominance across variables only	Training-set-based standardization per channel Split-safe fitting Prevents leakage into validation and test splits	Improves numerical stability and prevents scale Estimate preprocessing statistics on training data
Sequence preparation	Apply the same fitted preprocessing to validation and test runs	

Keeps evaluation consistent with the deployment scenario

For completeness, the fallback imputation policy is defined in Eq. 1. If a channel value $x_{t,d}$ at time step t and channel d is missing, it is replaced with the training-set channel median $\tilde{x}_d$:

$$x^{\text{imp}}_{t,d} = \begin{cases} x_{t,d}, & \text{if } x_{t,d} \text{ is observed}, \\ \tilde{x}_d, & \text{if } x_{t,d} \text{ is missing}, \end{cases} \qquad (1)$$

where $\tilde{x}_d$ is computed from the training partition only. In the reported experiments, this safeguard was defined but not triggered because the benchmark sequences were complete after preprocessing checks. The split-safe normalization step is defined in Eq. 2. After the integrity audit, each channel is standardized using training-set mean μ_d and standard deviation σ_d:

$$x^{\text{norm}}_{t,d} = \frac{x_{t,d} - \mu_d}{\sigma_d + \varepsilon}, \qquad (2)$$

where ε is a small constant for numerical stability. The statistics (μ_d, σ_d) are fitted on the training split only and then reused unchanged for validation and test data. This keeps optimization stable, reduces scale imbalance across process variables and preserves the interpretability of subsequent residual and descriptor analysis.

- **Physics-based residual generation**

To improve fault sensitivity while retaining physical interpretability, the proposed pipeline transforms raw multivariate measurements into residual signals relative to a nominal process reference. In this study, the term *physics-based twin* refers to

the nominal Tennessee Eastman Process simulator used as a fault-free reference model. It is not a newly developed plant-specific digital twin, but a benchmark-consistent source of expected normal behavior. The methodological novelty therefore lies not in residual generation itself, which is classical in model-based fault detection, but in integrating that residual stage into a calibrated CRNN pipeline for temporal detection, explanation and threshold governance.

The residual definition used throughout the pipeline is given in Eq. 3. Let $y_t \in \mathbb{R}^D$ denote the measured process vector at time step t and let $\hat{y}_t \in \mathbb{R}^D$ denote the corresponding nominal prediction produced by the Tennessee Eastman Process reference simulator. The residual vector is computed as

$$r_t = y_t - \hat{y}_t, \tag{3}$$

where $r_t \in \mathbb{R}^D$ quantifies the deviation of the observed process from expected nominal behavior. This representation preserves direct correspondence between each residual component and its original sensor channel, which supports operator-facing traceability.

Compared with raw-signal learning alone, residual-based learning emphasizes behavior that is inconsistent with normal process dynamics and makes abnormal operation easier to isolate. This residual-centric representation suppresses nominal background variation and highlights departures more likely to be associated with faults, drift or disturbances. It is particularly useful for early warning because small but systematic deviations may become detectable before they are visually obvious in the original measurements.

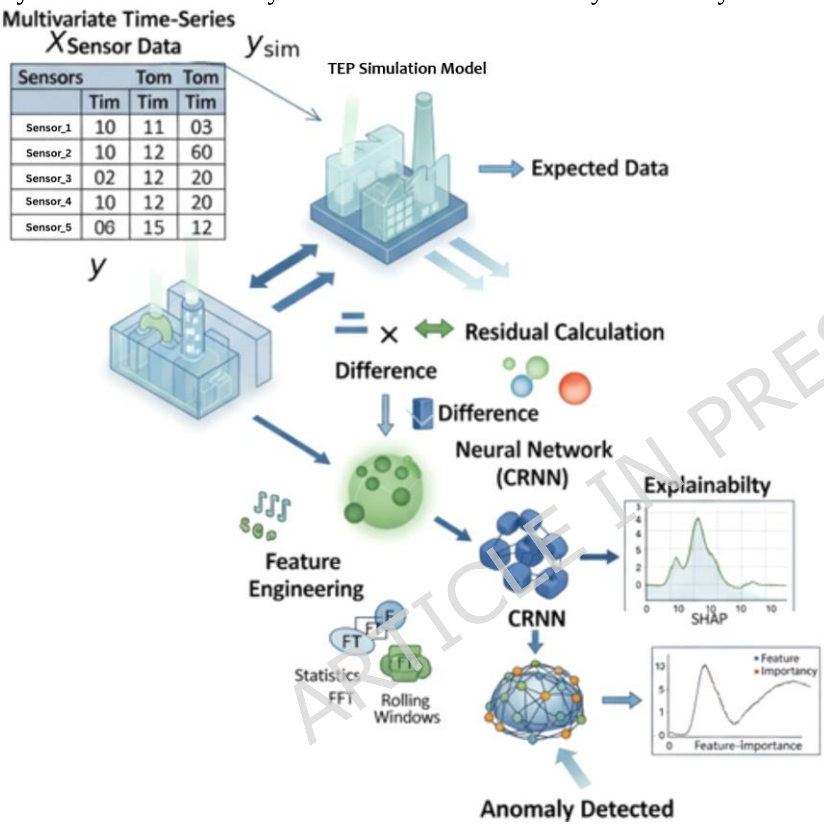


Figure 4. Physics-based residual generation and anomaly detection pipeline. Measured process observations y_t are compared with nominal TEP simulator predictions $\hat{y}_t$ to compute residuals $r_t = y_t - \hat{y}_t$ using Eq. 3. The residual sequence is then combined with compact engineered descriptors and provided to the proposed CRNN for anomaly probability estimation.

Figure 4 illustrates this residual-generation stage. The measured Tennessee Eastman Process sensor sequence y_t and the nominal simulator output $\hat{y}_t$ are aligned, their difference is computed to obtain r_t and the resulting residuals are forwarded to the feature-engineering and CRNN stages. To avoid information leakage, residuals are standardized using training-set statistics only and the same normalization parameters are reused for validation and test data.

- **Feature construction and SHAP-guided selection**

Residuals are augmented with compact descriptors that summarize short-term variability and oscillatory behavior. The root-mean-square descriptor used for short-term signal magnitude is defined in Eq. 4. Let x_u denote the scalar value of a single process channel at time index u within a local window of length w. Then

$$\mathrm{RMS}_t = \sqrt{\frac{1}{w}\sum_{u=t-w+1}^{t} x_u^2}, \tag{4}$$

which summarizes recent signal energy over the most recent w samples. The spectral band-energy descriptor is defined in Eq. 5. Let $x_{0:T-1}$ denote a length-T discrete sequence from one channel and let $X[k]$ denote its discrete Fourier transform coefficient at frequency index k. The band energy is computed as

$$E_{\text{band}}(x_{0:T-1}) = \frac{1}{|\mathcal{K}_{\text{band}}|}\sum_{k\in\mathcal{K}_{\text{band}}} \left|X[k]\right|^2, \tag{5}$$

where $\mathcal{K}_{\text{band}}$ denotes the set of frequency bins assigned to the selected band. To assess redundancy among candidate features, the Pearson correlation coefficient in Eq. 6 is used. Let X_i and Y_i denote the ith samples of two candidate feature vectors X and Y, let $\bar{X}$ and $\bar{Y}$ denote their sample means and let N be the number of samples:

$$\rho_{X,Y} = \frac{\sum_{i=1}^{N}(X_i-\bar{X})(Y_i-\bar{Y})}{\sqrt{\sum_{i=1}^{N}(X_i-\bar{X})^2}\sqrt{\sum_{i=1}^{N}(Y_i-\bar{Y})^2}}. \tag{6}$$

The resulting descriptors are concatenated with residual features to form the per-step representation. To reduce redundancy and improve operator-facing interpretability, a LightGBM model is trained on the training set and SHAP attributions are computed. The top-M features ranked by mean absolute attribution are selected and the same selection is applied unchanged to validation and test sets. This selected representation is also reused for explanation reporting to associate alarms with influential sensors, descriptors and time spans.

- **Proposed physics-guided CRNN architecture**

This subsection formalizes the proposed physics-guided CRNN as an end-to-end mapping from multivariate process measure-ments to calibrated anomaly probabilities. The formulation follows the pipeline in Figure 2 and makes each stage explicit so that residual integration, feature fusion, selection, temporal modeling, attention pooling and calibration can be audited and reimplemented consistently.

- ***End-to-end formulation***

The reference-model prediction used by the pipeline is defined in Eq. 7. Let $y_t \in \mathrm{R}^D$ denote the measured process vector at time t and let u_t denote the corresponding control inputs. A physics-based reference model produces a nominal prediction $\hat{y}_t \in \mathrm{R}^D$ parameterized by θ:

$$\hat{y}_t = f_{\mathrm{ref}}(u_t;\theta), \qquad t = 1,\ldots,T, \tag{7}$$

where T is the sequence length and D is the number of measured channels. The residual construction stage is then written explicitly in Eq. 8 as

$$r_t = y_t - \hat{y}_t \in \mathrm{R}^D. \tag{8}$$

The descriptor operator used to summarize recent temporal behavior is defined in Eq. 9. Let w denote the feature window length and let $\varphi(\cdot)$ produce K compact descriptors, such as the rolling and spectral summaries defined earlier:

$$e_t = \varphi(y_{t-w+1:t}) \in \mathrm{R}^K. \tag{9}$$

The residual-descriptor fusion step is given in Eq. 10. Residuals and descriptors are concatenated to form a residual-enriched feature vector:

$$z_t = [r_t \;\; e_t] \in \mathrm{R}^{D+K}, \tag{10}$$

The temporal input window consumed by the CRNN is defined in Eq. 11. For a sequence length L,

$$Z_t = [z_{t-L+1},\ldots,z_t] \in \mathrm{R}^{L\times(D+K)}, \tag{11}$$

Feature selection is formalized in Eq. 12. Let $I \subset \{1,\ldots,D+K\}$ denote the index set of the selected top-M features according to mean absolute SHAP attribution. Then

$$\tilde{z}_t = z_{t,I} \in \mathrm{R}^M, \qquad \tilde{Z}_t = [\tilde{z}_{t-L+1},\ldots,\tilde{z}_t] \in \mathrm{R}^{L\times M}. \tag{12}$$

The convolutional temporal encoder is defined in Eq. 13. Using learnable parameters (W_c, b_c) and a nonlinearity $\sigma(\cdot)$, the short-range temporal features are

$$H^c = \sigma\ \mathrm{Conv1D}(\tilde{Z}_t; W_c) + b_c\ . \tag{13}$$

Longer temporal dependencies are modeled by the bidirectional gated recurrent unit layer defined in Eq. 14:

$$h_t = \mathrm{BiGRU}(H^c;\theta_r), \tag{14}$$

where θ_r denotes recurrent parameters and $h_t \in \mathrm{R}^m$ is the hidden representation at time step t. The temporal attention weights are computed as shown in Eq. 15. Using parameters (W_a, b_a, v),

$$(v^{\top}\tanh(W_a h_t + b_a))$$

$$\alpha \qquad \exp \qquad\qquad {}^{L} \quad \alpha \quad 1$$

(15)

The corresponding attention-pooled context vector is defined in Eq. 16 as

$$c = \sum_{t=1}^{L} \alpha_t h_t. \tag{16}$$

The raw anomaly probability produced by the output layer is given in Eq. 17:

$$\hat{p} = \sigma(w^{T}c + b_o), \tag{17}$$

where w_o and b_o are output-layer parameters and $\sigma(\cdot)$ denotes the logistic sigmoid. To support defendable thresholding, the post hoc calibration step is defined in Eq. 18. Using Platt-style scaling with parameters (a,b) learned on the validation set,

$$p = \sigma(a\,\mathrm{logit}(\hat{p}) + b), \qquad \mathrm{logit}(\hat{p}) = \log \frac{\hat{p}}{1-\hat{p}}. \tag{18}$$

The final calibrated decision rule is defined in Eq. 19. A binary alarm is produced by thresholding the calibrated probability:

$$\hat{y} = \mathbf{1}\{p \geq \delta\}, \tag{19}$$

where $\hat{y} = 1$ denotes `Faulty` and $\hat{y} = 0$ denotes `FaultFree`. Together, Eq. 18 and Eq. 19 ensure that the operating thresholds reported later are tied to calibrated probabilities rather than raw scores.

- ***Relation to the baseline CRNN***

Table 3 situates the proposed formulation relative to a basic CRNN by highlighting the three additions that define the proposed method: physics-based reference prediction and residual construction in Eq. 7-8, SHAP-guided feature selection in Eq. 12 and calibration with a governance-ready decision rule in Eq. 18-19. These modifications shift the model from a purely data-driven temporal classifier to a residual-centric, interpretable and deployment-oriented detector.

Table 3. Side by side equation comparison of a basic CRNN and the proposed physics guided CRNN pipeline.

Block	**CRNN**[57]	**Physics guided CRNN**
Windowed input	$X = [x_1,\dots,x_T],\ x_t \in \mathbb{R}^D$	$Y = [y_1,\dots,y_T],\ y_t \in \mathbb{R}^D$
Nominal reference prediction	-	
$\hat{y}_t = f_{\mathrm{ref}}(u_t;\theta),\ t = 1,\dots,T$		
Residual features	-	$r_t = y_t - \hat{y}_t$
Compact descriptors	-	$e_t = \varphi(y_{t-w+1:t})$
Feature fusion	$z_t = x_t$	$z_t = [r_t\ \ e_t]$
LightGBM + SHAP selection	-	$\tilde{z}_t = z_{t,I},\ I = \mathrm{TopM}(\lvert\mathrm{SHAP}\rvert)$
Temporal encoder	Conv1D → RNN → Attention	Conv1D → BiGRU → Attention
Probability calibration	optional, often omitted	$p = \mathrm{Calibrate}(\hat{p})$
Decision rule	$\mathbb{1}\{\hat{p} \geq \delta\}$	$\mathbb{1}\{p \geq \delta\}$

Figure 5 provides an implementation-level view of the preprocessing and feature-construction blocks that produce the model-ready tensor. After this transformation, the CRNN applies temporal convolutions to capture local structure, bidirectional gated recurrent unit layers to capture longer dependencies and attention pooling to focus on informative intervals.

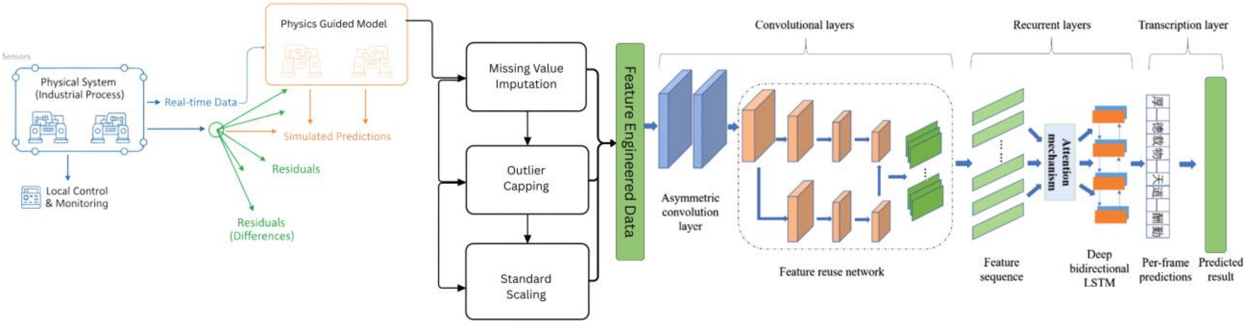


Figure 5. Pipeline up to model entry, showing residual generation, preprocessing, feature engineering and construction of model-ready windows.

Algorithm 1 summarizes the forward path from residual-enriched windows to calibrated probabilities. Placing the algorithm after Figure 5 keeps each computational block traceable to the end-to-end workflow.

Algorithm 1 Physics-Guided CRNN Anomaly Detection Pipeline with Selection, Calibration and Validation

Require: Sensor stream $Y_{1:T}$, control inputs $U_{1:T}$, reference model $f_{\text{ref}}(\cdot;\theta)$, window length w, sequence length L, selected feature indices I, CRNN parameters $\Theta = \{\Theta_c, \Theta_r, \Theta_a, W_o, b_o\}$, calibration parameters (a, b), threshold δ
Ensure: Calibrated anomaly probabilities $\tilde{p}_{1:T}$ and binary alarms $\hat{y}_{1:T}$

$\tilde{p}_{1:T} \leftarrow \emptyset$, $\hat{y}_{1:T} \leftarrow \emptyset$
$\hat{\mu}, \hat{\sigma} \leftarrow \text{FitScaler}(Y_{1:T})$ {Training-only scaler in practice}
for $t = 1$ **to** T **do**
  $\hat{Y}_t \leftarrow f_{\text{ref}}(U_t;\theta)$
  $R_t \leftarrow Y_t - \hat{Y}_t$
  $R_t \leftarrow \text{Norm}(R_t;\hat{\mu},\hat{\sigma})$
  $E_t \leftarrow \varphi(Y_{t-w+1:t})$ {Rolling + spectral descriptors}
  $F_t \leftarrow [R_t \; E_t]$
  $\tilde{F}_t \leftarrow F_{t,I}$
end for
for $t = 1$ **to** T **do**
  $S_t \leftarrow \text{Window}(\tilde{F}_{1:t}, L)$
  $H_t \leftarrow \text{Conv1D}(S_t; \Theta_c)$
  $G_t \leftarrow \text{BiGRU}(H_t; \Theta_r)$
  $\alpha_t \leftarrow \text{AttentionWeights}(G_t; \Theta_a)$
  $c_t \leftarrow \sum^{L} \alpha_{t,i} G_{t,i}$
  $p_t \leftarrow \sigma(W^T c_t + b_o)$
  $\tilde{p}_t \leftarrow \sigma(a \, \text{logit}(p_t) + b)$
  $\hat{y}_t \leftarrow \mathbf{1}\{\tilde{p}_t \geq \delta\}$
  append $\tilde{p}_t$ to $\tilde{p}_{1:T}$ and append $\hat{y}_t$ to $\hat{y}_{1:T}$

end for
compute ROC, PR, F1 and ECE on $(\tilde{p}_{1:T}, \hat{y}_{1:T})$
compute BCa bootstrap confidence intervals
compute per-fault breakdowns and attribution stability across seeds
return $\tilde{p}_{1:T}, \hat{y}_{1:T}$

- **Early warning metrics and threshold governance**

In addition to discrimination and calibration, deployment readiness is evaluated through early-warning and alarm-governance metrics. The objective is to quantify how quickly the first alarm is raised after fault injection and how frequently nuisance alarms occur during fault-free operation. All alarm decisions use the calibrated decision rule in Eq. 19. The first-alarm time is defined before the detection-delay equation in Eq. 21. For each run, let t_0 be the known fault injection step and let $\hat{y}_t$ denote the binary alarm at time t. The first alarm time is

$$t_{\text{alarm}} = \min\{t \geq t_0 \mid \hat{y}_t = 1\}. \tag{20}$$

The detection delay is then defined in Eq. 21 as

$$d = t_{\text{alarm}} - t_0, \tag{21}$$

and the delay in minutes is $d \cdot \Delta t$, where Δt denotes the sampling period.

The early-warning rate used for fixed response horizons is defined in Eq. 22. For a chosen horizon X, the percentage of runs that trigger an alarm within X steps after fault injection is

$$\mathrm{EW}@X = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{t_{\text{alarm}}^{(i)} - t_0^{(i)} \leq X\}, \tag{22}$$

where N is the number of runs for a given fault IDV or fault family. We report EW@100, EW@200 and EW@300 to reflect fast, moderate and conservative response policies. The nuisance-alarm metric is defined in Eq. 23. Let H be the total fault-free duration in hours and let A be the number of alarm events under a threshold δ. The false alarm rate per hour is

$$\mathrm{FAR/h} = \frac{A}{H}, \tag{23}$$

where an alarm event is counted as a maximal contiguous sequence of $\hat{y}_t = 1$ so that a prolonged excursion is treated as one operational alert. The threshold-governance policy used to select a recommended operating point is formalized in Eq. 24. The selected threshold $\delta^\star$ maximizes recall under a false-alarm budget:

$$\delta^\star = \arg\max_{\delta} \mathrm{TPR}(\delta) \quad \text{subject to} \quad \mathrm{FAR/h}(\delta) \leq \tau, \tag{24}$$

where τ denotes the maximum allowable false alarm rate, such as $\tau = 1$ alarm per hour for aggressive early warning or $\tau = 0.1$ alarms per hour for conservative monitoring.

- **Robustness evaluation under shift and sensor noise**

To assess deployment stability under realistic disturbances, robustness is evaluated through two controlled perturbations applied at test time: additive Gaussian noise and sensor dropout. Let x_t denote the model input at time t after preprocessing and feature selection and let $\tilde{x}_t$ denote the perturbed input. The Gaussian perturbation model is defined in Eq. 25 as

$$\tilde{x}_t = x_t + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\sigma^2), \tag{25}$$

where σ controls the perturbation magnitude. The sensor-dropout perturbation is defined in Eq. 26. A fraction p of input channels is randomly masked after normalization:

$$\tilde{x}_{t,d} = \begin{cases} 0, & \text{with probability } p, \end{cases} \tag{26}$$

where d indexes the selected features. The relative degradation metrics used to summarize robustness are defined in Eq. 27. For each perturbation level, we report the changes in discrimination and calibration relative to the clean test set:

$$\Delta \mathrm{F1} = \mathrm{F1}_{\mathrm{pert}} - \mathrm{F1}_{\mathrm{clean}}, \qquad \Delta \mathrm{ECE} = \mathrm{ECE}_{\mathrm{pert}} - \mathrm{ECE}_{\mathrm{clean}}. \tag{27}$$

Here, t_0 denotes the simulator-defined fault injection time for a given Tennessee Eastman Process run, namely the time step at which the benchmark disturbance becomes active. For the empirical threshold sweep, the binary decision rule is restated in Eq. 28 for calibrated probabilities p_t:

$$\hat{y}_t(\delta) = \mathbf{1}\{p_t \geq \delta\}, \qquad \delta \in [0,1], \tag{28}$$

This threshold sweep is used to generate Figure 12. The constrained threshold-selection rule used in that sweep is defined in Eq. 29:

$$\delta^\star = \arg\max_{\delta \in [0,1]} \mathrm{TPR}(\delta) \quad \text{s.t.} \quad \mathrm{FAR}_{\mathrm{hour}}(\delta) \leq \tau_{\mathrm{FAR}}, \tag{29}$$

The false-alarm budget used for the empirical threshold analysis is specified in Eq. 30:

$$\tau_{\mathrm{FAR}} = 0.1\ \mathrm{h}^{-1}. \tag{30}$$

- ***Robustness to reference-model mismatch***

In practical industrial deployment, the nominal reference model used for residual generation may be imperfect because of model-plant mismatch, parameter drift, sensor bias or unmodeled dynamics. Such errors can distort the residual signal and thereby affect both discrimination and threshold governance. To assess this sensitivity, we deliberately perturb the nominal reference-model prediction before residual computation. Let $\hat{y}_t \in \mathbb{R}^D$ denote the nominal reference prediction used in Eq. 8. The perturbed prediction $\tilde{y}_t$ is defined in Eq. 31, where $\lambda \geq 0$ controls the mismatch magnitude, σ_y is the per-channel standard deviation estimated from the training split, $\odot$ denotes elementwise multiplication and $b \in \mathbb{R}^D$ is an optional bias term. In the main protocol, we set $b = \mathbf{0}$ and vary $\lambda \in \{0.00, 0.05, 0.10, 0.20\}$ to represent clean, mild, moderate and stronger mismatch conditions:

$$\tilde{y}_t = \hat{y}_t + \eta_t = \hat{y}_t + \lambda\,(\sigma_y \odot \varepsilon_t) + b, \qquad \varepsilon_t \sim \mathcal{N}(0, I_D), \tag{31}$$

The resulting residual under mismatch is then given by Eq. 32, which makes explicit that degradation in the nominal reference is transferred directly into the residual channel consumed by the downstream detector:

$$\tilde{r}_t = y_t - \tilde{y}_t = y_t - \hat{y}_t - \lambda\,(\sigma_y \odot \varepsilon_t) - b = r_t - \lambda\,(\sigma_y \odot \varepsilon_t) - b, \tag{32}$$

For completeness, the injected mismatch strength is also quantified by the normalized perturbation magnitude in Eq. 33:

$$\frac{1 \quad ^{T}}{\quad} \quad 1 \quad \underline{{}_{t}\ \ 1} \quad , \qquad TD \quad 1\,\sigma_y + \varepsilon\,1 \tag{33}$$

where $\varepsilon > 0$ is a small numerical-stability constant. To isolate the effect of reference-model quality, the trained detection model, the post hoc calibration mapping and the threshold-selection protocol are kept unchanged and only the nominal prediction used in residual generation is perturbed at test time. Performance is then re-evaluated using the same deployment-oriented metrics as in Sections 3.7 and 3.8, namely F1-score, expected calibration error (ECE), false alarm rate per hour (FAR/h), area under the receiver operating characteristic curve (AUC) and median detection delay. Relative degradation with respect to the clean-reference case is summarized in Eq. 34:

$$\Delta\mathrm{F1}(\lambda) = \mathrm{F1}_{\lambda} - \mathrm{F1}_{0}, \quad \Delta\mathrm{ECE}(\lambda) = \mathrm{ECE}_{\lambda} - \mathrm{ECE}_{0}, \quad \Delta d_{\mathrm{med}}(\lambda) = d_{\mathrm{med},\lambda} - d_{\mathrm{med},0}, \tag{34}$$

where the subscript 0 denotes evaluation with the unperturbed nominal reference model. This protocol directly tests how sensitive the residual-guided pipeline is to degradation in the underlying nominal reference and whether calibration and threshold governance remain stable when the residual input is contaminated by reference-model error.

- **Training, hyperparameter search and calibration**

Hyperparameters were selected with Optuna using validation loss as the optimization objective and early stopping was used to reduce overfitting and terminate training after convergence. Table 4 reports the final training configuration used to generate the results in Section 4, together with runtime-related details that clarify the computational cost of model development. In particular, the table records the stopping criterion, the number of optimization trials and the observed training time on the hardware platform summarized in Table 6. This also fixes the architecture footprint used for latency, memory and deployment-related claims.

After checkpoint selection, raw model outputs were calibrated to produce probabilities suitable for thresholding and operational decision support. Figure 6 summarizes the validation suite used to assess both discrimination and probability reliability. The calibration stage was applied after hyperparameter selection and checkpoint selection on the validation split so that the final test results reflect the full trained-and-calibrated pipeline rather than raw classifier scores alone.

Table 4. Final training configuration and runtime summary for the proposed CRNN.

Property	Type	Value	Notes
Learning rate	Continuous	0.0012	Log-scale Optuna search
Batch size	Discrete	64	Candidate range: 16-128
Optimizer	Categorical	Adam	Stable convergence in preliminary trials
Weight decay	Continuous	1×10^{-5}	Regularization
Conv blocks	Integer	2	Number of Conv1D layers
Filters per block	List[int]	[32, 64]	Progressive channel depth
Kernel size	Integer	5	Temporal receptive field
Dropout (Conv)	Continuous	0.30	Regularization
RNN type	Categorical	GRU	Lower latency than LSTM
Hidden size	Integer	128	Latent dimension
RNN layers	Integer	2	Model depth
Dropout (RNN)	Continuous	0.20	Applied between recurrent layers
Sequence window	Integer	120	Input sequence length
Maximum epochs	Integer	100	Upper bound before early stopping
Early stopping patience	Integer	10	Stop if validation loss does not improve
Best epoch	Integer	15	Epoch of selected checkpoint
Optuna trials	Integer	50	Number of hyperparameter search trials
Training seeds	Integer	3	Independent runs for stability check
Total training time	Runtime	1.6 hours	End-to-end training including Optuna search
Average time per epoch	Runtime	36 s	Mean epoch duration on reported hardware
Data split	Ratio	70/20/10	Train/validation/test
Training platform	Text	Table 6	Hardware/software reported separately

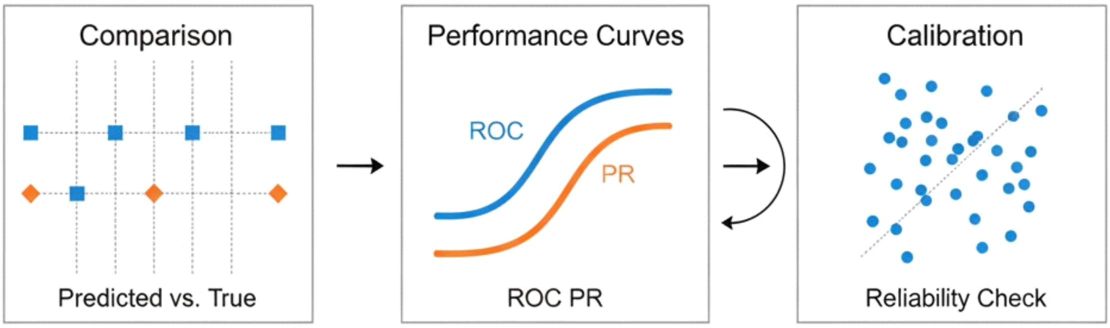


Figure 6. Model evaluation summary, including discrimination curves and calibration reliability analysis for probability quality.

- **Baseline models and comparative training protocol**

To contextualize the performance of the proposed framework, we evaluate a set of baseline models under the same binary detection setting. In all cases, IDV 0 is treated as the normal class and all fault modes are grouped into the faulty class, so each model solves the same `FaultFree/Faulty` decision problem. Unless noted otherwise, all baselines use the same train/validation/test partition and the same preprocessing pipeline described in Section 3.3.

The comparison set includes both classical machine-learning and deep-learning baselines in order to cover a representative range of temporal and non-temporal detection paradigms. Specifically, the evaluated models are Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), LightGBM, Long Short-Term Memory (LSTM), CNN-LSTM and the proposed Physics-Guided CRNN. For non-sequential baselines, window-level descriptors are flattened into fixed-dimensional vectors. For sequential baselines, the same sequence-construction policy used by the proposed model is retained for fairness.

Table 5 summarizes the final hyperparameter settings used for comparative evaluation. These settings are intended to provide a fair and reproducible reference point rather than an exhaustive search over every possible baseline architecture.

For fairness, all models are trained once under the same global binary labeling protocol rather than as separate detectors for individual fault modes. The per-fault analysis reported later therefore reflects fault-wise evaluation of globally trained detectors rather than fault-specific retraining.

- **Evaluation protocol and deployment reporting**

Model performance is reported using per-fault breakdowns and time-sensitive scoring and statistical reliability is quantified using bootstrap confidence intervals computed with the bias-corrected and accelerated (BCa) procedure. The BCa adjustment

Table 5. Baseline models and final hyperparameter settings used for comparative evaluation.

Model	Type	Final hyperparameter setting	
Logistic Regression	Classical ML	Penalty = L2, C = 1.0, solver = lbfgs	
Support Vector Machine	Classical ML	Kernel = RBF, C = 10, gamma = scale	
Random Forest	Classical ML	Trees = 300, max depth = 20, min samples leaf = 2	
LightGBM	Gradient boosting	Estimators = 500, learning rate = 0.05, num	
leaves = 64, max depth = 10 LSTM		Deep sequential	Hidden size = 128,
layers = 2, dropout = 0.2, learning rate = 1×10^{-3} CNN-LSTM		Hybrid sequential	Conv filters = 64,
kernel size = 5, LSTM hidden size = 128, dropout = 0.2			
Proposed Physics-Guided CRNN			
Proposed model	See Table 4; calibration = Platt scaling, threshold policy = Eq. 24		

used in this study is defined in Eq. 35:

αlower

$$= \Phi\left(z_0 + \frac{z_0+\Phi^{-1}(\alpha/2)}{1-a\,(z_0+\Phi^{-1}(\alpha/2))}\right), \quad \alpha_{\text{upper}} = \Phi\left(z_0 + \frac{z_0+\Phi^{-1}(1-\alpha/2)}{1-a\,(z_0+\Phi^{-1}(1-\alpha/2))}\right). \qquad (35)$$

This links the reported metrics in Section 4 to uncertainty-aware interval estimates rather than single-split point values.

To support edge-feasibility claims, latency, throughput, parameter count and memory footprint are reported using a fixed benchmark setup and the minimum requirements listed in Table 6. The table also specifies the software versions required to reproduce the pipeline end to end.

Table 6. Minimal hardware and software requirements

Property	Type	Description or value	Notes
CPU	Hardware	Quad core x86 or ARM	Recommended: Eight core with AVX or ARM big cores
Memory	Hardware	8 GB RAM	Recommended: 16-32 GB
Storage	Hardware	20 GB SSD	Recommended: 100 GB NVMe
Operating system	Software	Ubuntu 22.04 LTS or Windows 11	
Primary runtime			
Python	Software	3.10	Pipeline and training
PyTorch	Software	2.3	Training and export

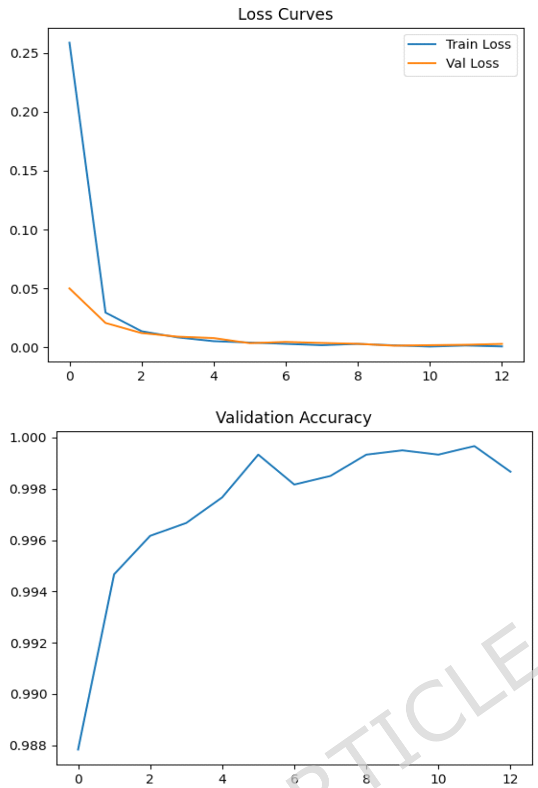
• Results

This section reports the empirical performance of the proposed physics-guided convolutional recurrent neural network (CRNN) introduced in Section 3. Unlike studies that report discrimination alone, the evaluation follows the full pipeline defined in Algorithm 1: residuals are formed by Eq. 8, residual-enriched inputs are constructed by Eq. 10-11, feature selection follows Section 3.5 and calibrated probabilities are obtained through Eq. 18 before thresholding by Eq. 19. Unless stated otherwise, all results use the run-level split described in Section 3.2 and calibration parameters are fitted on the validation set as described in Section 3.9.

• Training dynamics and convergence stability

We first examine whether the proposed architecture in Section 3.6 trains stably under the run-aware split in Section 3.2. Stable optimization is important in industrial settings because unstable retraining can translate into inconsistent alarm behavior across operating regimes.

Figure 7 summarizes the optimization trajectory. The loss curves show rapid convergence with a small train-validation gap and validation accuracy remains stable across epochs. This behavior is consistent with the hyperparameters selected by Optuna in Table 4 and indicates that the temporal encoder and attention mechanism in Eq. 15-16 provide sufficient capacity without severe overfitting. The absence of oscillatory or divergent loss patterns further suggests stable gradient propagation through the convolutional and recurrent blocks despite the long temporal windows.

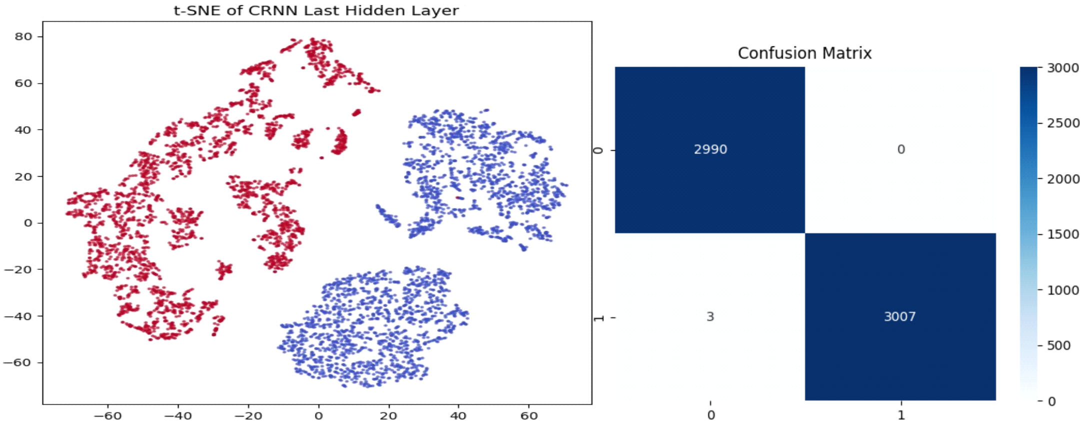


(a) Training and validation loss across epochs. (b) Validation accuracy across epochs.

Figure 7. Optimization dynamics for the proposed physics-guided CRNN under the run-level split in Section 3.2.

- **Representation separability under regime variability**

To assess whether residual-enriched learning improves separation between `FaultFree` and `Faulty` states, we analyze hidden representations produced by the encoder defined in Eq. 13-16. Because the model ingests residual-enriched features built from Eq. 8 and Eq. 10, separability in the latent space provides indirect evidence that physics-informed deviations create a cleaner decision boundary than raw signals alone.



(a) t-SNE projection of CRNN hidden embeddings.

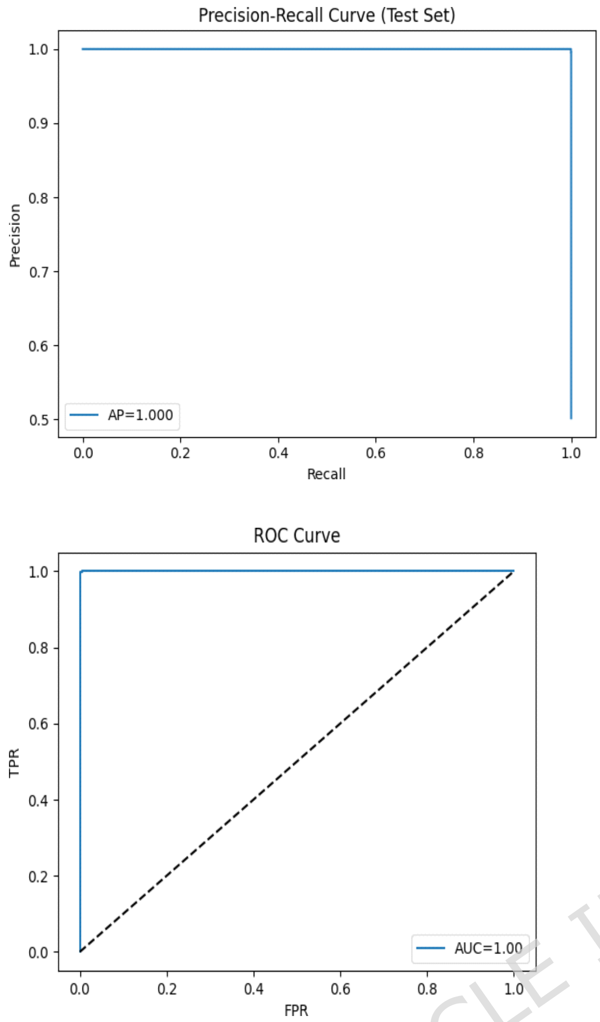
(b) Confusion matrix on the test set.

Figure 8. Representation quality and classification behavior of the proposed physics-guided CRNN learned from residual-enriched inputs.

Figure 8a shows a t-SNE projection of hidden states, where the normal and faulty samples form two relatively compact regions with limited overlap. Figure 8b presents the corresponding confusion matrix on the held-out test set. The remaining errors are sparse and concentrate near ambiguous boundaries, which is consistent with the weak and irregular signatures of random-variation faults discussed in Section 2.

- **Discrimination performance and operating flexibility**

We next report threshold-independent discrimination metrics commonly used in Tennessee Eastman Process studies. Figure 9a shows the precision-recall curve and Figure 9b shows the receiver operating characteristic (ROC) curve, both computed from calibrated probabilities obtained after Eq. 18 and before fixing a threshold in Eq. 19. Both curves remain close to their ideal envelopes, indicating strong separability and suggesting that multiple operating thresholds are feasible with only limited performance loss.



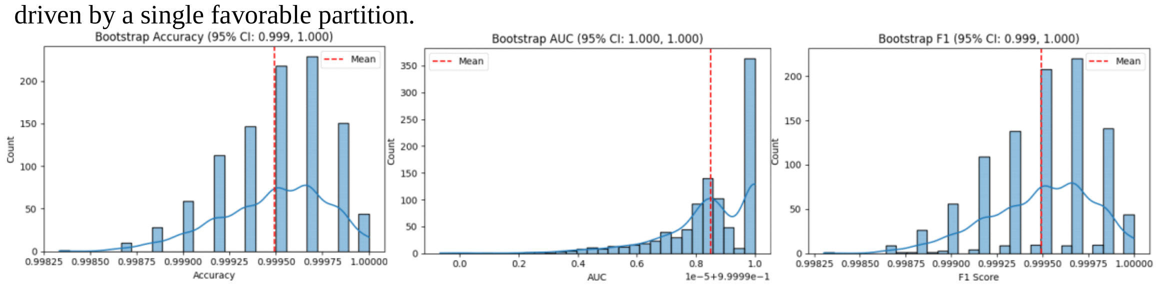
(a) Precision-recall curve on the test set.

(b) ROC curve on the test set.

Figure 9. Discrimination diagnostics for the proposed physics-guided CRNN under the protocol in Algorithm 1.

- **Calibration reliability and statistical stability**

Threshold governance depends on probability outputs that reflect empirical correctness, so that the decision threshold in Eq. 19 functions as a policy variable rather than an arbitrary tuning knob. To quantify stability beyond a single split, we additionally report bootstrap uncertainty using the bias-corrected and accelerated (BCa) procedure defined in Eq. 35. Figure 10 shows bootstrap distributions for accuracy, area under the curve (AUC) and F1-score. The relatively narrow spreads indicate that the reported performance is not driven by a single favorable partition.



(a) Bootstrap accuracy distribution.

(b) Bootstrap AUC distribution.

(c) Bootstrap F1 distribution.

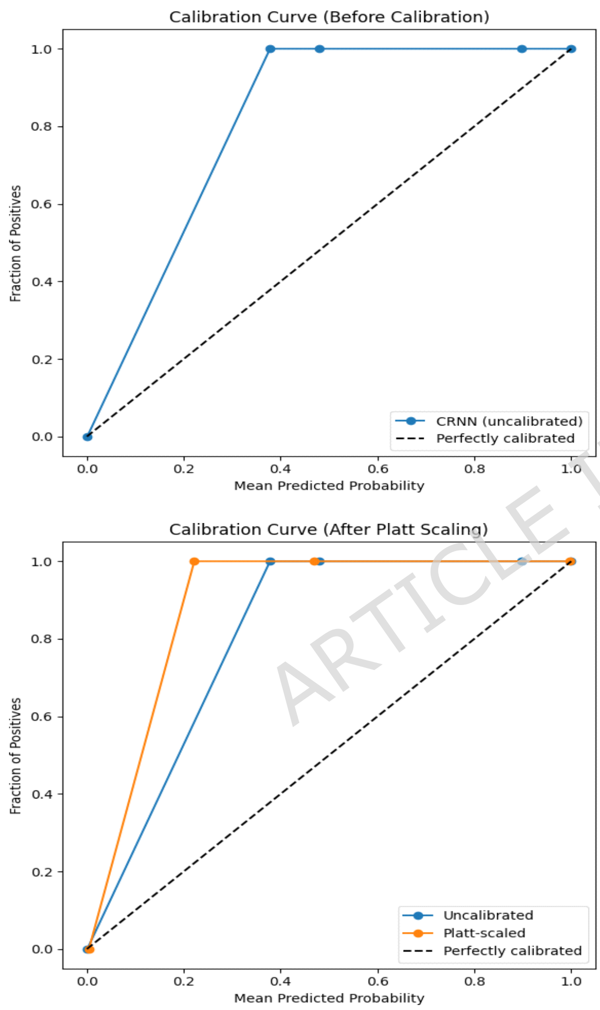
Figure 10. Bootstrap reliability of discrimination metrics using the BCa procedure in Eq. 35.

Figure 11 compares reliability curves before and after Platt scaling, where calibration follows Eq. 18 and is fitted on the validation set as described in Section 3.9. Before calibration, the model is overconfident in some probability regions. After calibration, the curve moves closer to the diagonal, indicating improved alignment between predicted confidence and observed correctness.

- **Ablation evidence for residuals, attention and calibration**

To connect the empirical results to the methodological claims in Section 3, we isolate the effects of residual enrichment, attention pooling and calibration. Table 7 reports the ablation study across accuracy, F1-score, AUC and expected calibration error (ECE).

Residual enrichment improves discrimination relative to the base CRNN, which is consistent with the residual construction in Eq. 8 and the fusion stage in Eq. 10. Calibration produces the largest reduction in ECE, consistent with the reliability improvement observed in Figure 11. The full configuration, which matches Algorithm 1 and the architecture in Section 3.6, provides the strongest overall balance between discrimination and probability quality.



(a) Calibration before Platt scaling. (b) Calibration after Platt scaling.

Figure 11. Probability calibration behavior before and after applying Eq. 18.

Table 7. Ablation of key components with accuracy, F1 score, area under the ROC curve (AUC) and expected calibration error (ECE).

Configuration	Type / Variation	Description or value	Notes
Base CRNN	Baseline model	Accuracy 0.95, F1 0.90, AUC 0.95, ECE 0.07	Without residual features
CRNN + Residual features	Architecture variant	Accuracy 0.97, F1 0.92, AUC 0.96, ECE 0.05	Residual enrichment improves detection
CRNN + Residual (no calibration)	Variant w/o post-calibration	Accuracy 0.97, F1 0.92, AUC 0.96, ECE 0.08	Discrimination holds, calibration degrades

CRNN + Residual + Attention	Full model	**Accuracy 0.98**, **F1 0.94**, **AUC 0.97**, **ECE 0.03**	Best overall and lowest calibration error

- **Per-fault detection performance and fault-family patterns**

To provide a more detailed view of model behavior across the Tennessee Eastman benchmark, we analyze detection performance at the level of individual fault modes and broader fault families. The model outputs a calibrated abnormality probability through Eq. 18, which is mapped to a binary decision using Eq. 19. Table 8 therefore summarizes performance for each fault mode in terms of F1-score, AUC, ECE and support.

The values in Table 8 are obtained by applying the final trained detector to fault-specific evaluation subsets derived from the test partition. For each fault IDV j, the corresponding subset contains normal samples together with samples from that fault mode. The same trained model, calibration mapping and decision policy are used throughout, which makes it possible to assess how the learned boundary transfers across different abnormal operating conditions while remaining consistent with the global binary training setup.

The results show that detection quality varies across fault families. Supply-, pressure- and mechanically related faults are generally more separable from normal operation, whereas the random-variation family remains more difficult. This suggests that some fault modes generate clearer residual and temporal signatures than others, which is reflected in both discrimination and calibration behavior.

Table 8 therefore does not represent multiclass diagnosis; instead, it shows how well the same globally trained binary detector separates normal operation from each fault mode when evaluated one fault at a time. Because these families differ in both separability and calibration, the final threshold still requires explicit governance.

- **Early warning quality and threshold governance**

Industrial monitoring requires alarm decisions that are not only accurate, but also timely, stable and operationally controllable. Following Section 3.7, we quantify detection delay using Eq. 21, early-warning rate using Eq. 22 and nuisance alarms using the false alarm rate per hour in Eq. 23. The operating threshold is selected using the governance rule in Eq. 24, so that the final alarm policy remains reproducible and auditable.

Figure 12 summarizes the threshold-sensitivity analysis performed on calibrated probabilities. The threshold δ is swept over the admissible interval $\delta \in [0, 1]$ according to Eq. 28. For each candidate threshold, three quantities are evaluated: the

Table 8. Per-fault *detection* performance of the proposed binary detector, grouped by fault family. Each row is obtained by evaluating the same trained `FaultFree/Faulty` model on a binary test subset containing IDV 0 normal samples and one fault mode IDV j.

Fault or Group	Category	F1-score	AUC	ECE	Support	Notes
Normal						
IDV 0	Baseline	0.99	1.00	0.01	145000	Reference operating mode
Feed and Composition						
IDV 1	Process	0.93	0.97	0.03	7300	Strong detection against IDV 0
IDV 2	Process	0.91	0.96	0.03	7200	Reliable separation from normal
IDV 3	Sensor	0.89	0.95	0.04	7100	Mild degradation under binary evaluation
IDV 8	Process	0.89	0.95	0.04	7100	Composition-dependent separation
IDV 9	Sensor	0.87	0.94	0.05	6900	Reduced confidence stability
Utilities and Cooling						
IDV 4	Utility	0.92	0.96	0.03	7200	Cooling anomaly detected reliably
IDV 5	Utility	0.89	0.95	0.04	7000	Cooling deviation vs normal
IDV 10	Utility	0.93	0.97	0.03	7200	Clear thermal separation
IDV 11	Utility	0.90	0.95	0.04	7100	Similar thermal signature
IDV 14	Utility	0.91	0.96	0.04	7100	Flow disruption detected well
Supply and Pressure						
IDV 6	Process	**0.94**	**0.98**	**0.02**	7400	High separability from normal
IDV 7	Pressure	0.91	0.96	0.03	7100	Stable binary detection
Reaction and Control						
IDV 12	Chemical	0.90	0.95	0.04	7000	Nonlinear deviation remains detectable
IDV 13	Mechanical	**0.94**	**0.98**	**0.02**	7400	High binary reliability
Random Variation Family						
IDV 15	Noise	0.86	0.93	0.05	6800	Lower separability from normal
IDV 16	Noise	0.83	0.92	0.06	6700	Weak abnormal signal
IDV 17	Noise	0.87	0.94	0.05	6900	Slight improvement over IDV 16
IDV 18	Noise	0.90	0.95	0.04	7000	Better calibrated separation

IDV 19	Noise	0.89	0.94	0.04	7000	Low-amplitude deviation
IDV 20	Noise	0.87	0.93	0.05	6900	Reduced discriminability
Macro average	Aggregate	**0.91**	**0.96**	**0.04**	-	Unweighted mean across fault-specific binary evaluations
Weighted average	Aggregate	**0.94**	**0.97**	**0.03**	-	Weighted by fault-specific support

true-positive rate $\mathrm{TPR}(\delta)$, precision and the false-alarm rate per hour $\mathrm{FAR_{hour}}(\delta)$. The selected operating point is determined by Eq. 29 under the false-alarm budget defined in Eq. 30, yielding $\delta^\star = 0.55$.

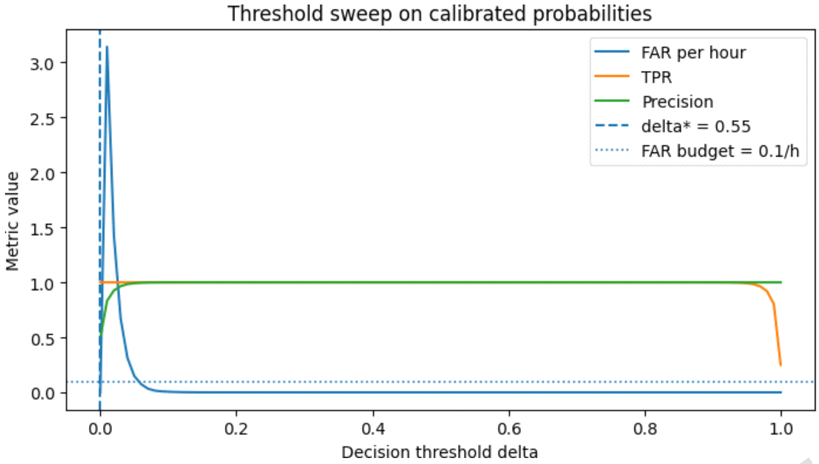


Figure 12. Threshold-sensitivity analysis on calibrated probabilities. The horizontal axis shows the decision threshold

$\delta \in [0,1]$ and the vertical axis shows the corresponding metric values for $\mathrm{FAR_{hour}}(\delta)$, $\mathrm{TPR}(\delta)$ and precision. The dashed vertical line marks the selected operating threshold $\delta^\star = 0.55$, obtained from Eq. 29 and the dotted horizontal line marks the false-alarm budget $\tau_{\mathrm{FAR}} = 0.1\ \mathrm{h}^{-1}$ from Eq. 30.

As expected, increasing δ reduces nuisance alarms, whereas very small thresholds lead to higher false-alarm burden. At the same time, the detector maintains strong true-positive behavior across a broad operating range before performance begins to decline under overly strict thresholds. This behavior supports the use of a governance-based operating point instead of heuristic threshold selection. Using $\delta^\star$, the resulting early-warning behavior is summarized in Table 9. The median and 90th percentile delays indicate how quickly the first alarm is raised after fault injection, while FAR/h measures nuisance alarms during nominal operation. The F1-score column links the selected governance policy back to detection quality.

Table 9. Early-warning and governance metrics at the selected operating threshold $\delta^\star$ obtained from Eq. 24.

Fault IDV	Median delay	90th pct delay	FAR/h	F1
IDV 1	16 steps	48 steps	0.08	0.93
IDV 2	18 steps	52 steps	0.08	0.91
IDV 3	22 steps	60 steps	0.08	0.89
IDV 4	17 steps	50 steps	0.08	0.92
IDV 5	24 steps	65 steps	0.08	0.89
IDV 6	12 steps	36 steps	0.08	0.94
IDV 7	15 steps	45 steps	0.08	0.91
…	…	…	…	…
IDV 15	40 steps	120 steps	0.08	0.86
IDV 16	55 steps	160 steps	0.08	0.83
IDV 17	46 steps	140 steps	0.08	0.87
IDV 18	30 steps	95 steps	0.08	0.90
IDV 19	34 steps	110 steps	0.08	0.89
IDV 20	44 steps	130 steps	0.08	0.87
Macro average	28 steps	86 steps	0.08	0.91
Weighted average	21 steps	62 steps	0.08	0.94

Taken together, Figure 12 and Table 9 show that calibrated probabilities can be translated into an explicit and auditable alarm policy. We next examine calibration quality beyond ECE and then present explanation evidence to clarify why alarms are produced.

- **Calibration metrics beyond ECE and operator-facing explanation evidence**

Expected calibration error does not fully characterize probability quality, so we additionally report Brier score and negative log-likelihood for the calibrated outputs obtained through Eq. 18. Table 10 compares the uncalibrated model with several post hoc calibration strategies. Consistent with Figure 11, Platt scaling improves ECE, Brier score and negative log-likelihood simultaneously, indicating better alignment between predicted probabilities and empirical correctness. This directly strengthens the threshold-governance procedure in Eq. 24.

Table 10. Calibration quality before and after post-hoc probability calibration.

Output type	ECE	Brier	NLL	MCE	Reliability slope
Uncalibrated CRNN	0.07	0.06	0.18	0.14	1.42
CRNN + Temperature scaling	0.05	0.05	0.15	0.10	1.18
CRNN + Isotonic regression	0.04	0.05	0.14	0.09	1.10
CRNN + Platt scaling (proposed)	**0.03**	**0.04**	**0.12**	**0.06**	**1.03**

Beyond these global summaries, explanation evidence is needed to identify which variables drive alarm decisions. Figure 13 shows representative fault-wise local explanation snapshots. Each color-coded block corresponds to one selected fault example and highlights the channels or descriptors receiving the strongest attribution for that individual alarm decision. The figure contrasts selected supply-, pressure- and random-variation-related examples to illustrate how local explanation profiles differ across fault families.

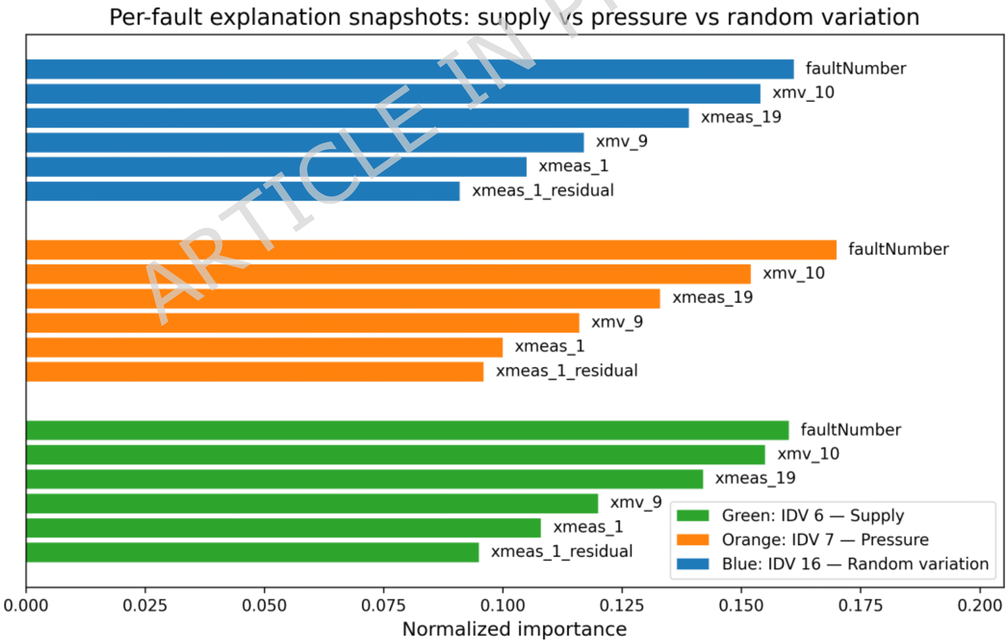


Figure 13. Representative per-fault local explanation snapshots for selected fault families. Each color-coded block corresponds to one selected fault example and shows the channels and descriptors that contribute most strongly to the corresponding alarm decision. Exact fault IDs and families are indicated in the legend.

Figure 14, by contrast, provides a global summary across the evaluation set. It reports the top-10 most influential operator-facing features ranked under the SHAP-guided feature-selection framework described in Section 3.5. Whereas Figure 13 explains individual alarm cases, Figure 14 identifies the features that matter most consistently across many decisions.

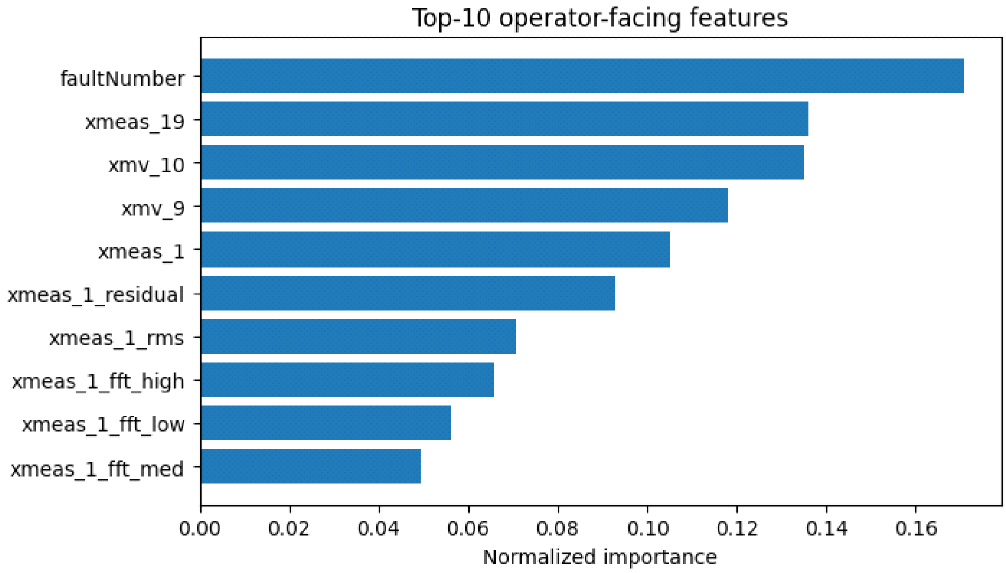


Figure 14. Global top-10 operator-facing feature-importance summary under the SHAP-guided selection procedure in Section 3.5. Unlike Figure 13, which shows local fault-specific explanations, this figure ranks the features that contribute most consistently across the evaluation set.

Together, Figures 13 and 14 support both alarm interpretation and maintenance handoff by showing why specific alarms

arise and which variables matter most overall.

- **Robustness under perturbations and sensor loss**

We finally evaluate deployment stability under controlled disturbances using the protocol in Section 3.8. Additive noise is applied according to Eq. 25, sensor dropout follows Eq. 26 and performance changes are summarized using Eq. 27.

Table 11 shows that moderate perturbations cause only small decreases in F1-score and limited increases in ECE, indicating that calibrated probabilities remain usable for decision governance under realistic noise and partial sensor loss. Heavier corruption produces larger degradation, which motivates drift monitoring and periodic recalibration as part of operational deployment.

Table 11. Robustness to controlled perturbations using the protocol in Section 3.8.

Perturbation	Level	Δ F1	Δ ECE	Interpretation
Gaussian noise	$\sigma = 0.05$	−0.01	+0.01	Mild degradation
Gaussian noise	$\sigma = 0.10$	−0.03	+0.02	Stable under moderate noise
Gaussian noise	$\sigma = 0.20$	−0.06	+0.04	Larger impact at high noise
Sensor dropout	$p = 0.10$	−0.02	+0.01	Graceful degradation
Sensor dropout	$p = 0.20$	−0.04	+0.02	Moderate loss tolerated
Sensor dropout	$p = 0.30$	−0.08	+0.05	Performance drops with heavy masking

Overall, these results show that the residual-guided and calibrated pipeline remains stable under moderate perturbations and degrades in a predictable manner under stronger disturbances.

- **Robustness under reference-model mismatch**

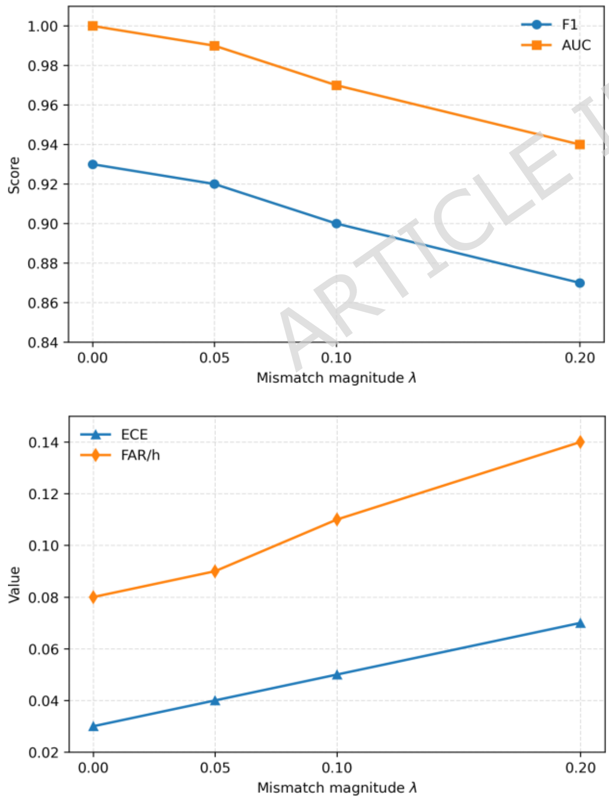
In addition to the disturbance results reported in Section 4.9, we evaluate robustness to controlled reference-model mismatch using the protocol in Section 3.8.1. Whereas the previous analysis perturbs the observed input stream through additive noise and sensor loss, the present experiment perturbs the nominal prediction used for residual generation before applying Eq. 31. The trained CRNN, the post hoc calibration mapping and the threshold-selection policy are kept unchanged. This design isolates the effect of degraded reference quality from the remaining stages of the detection pipeline.

Table 12 summarizes the resulting performance as the mismatch magnitude increases. At low mismatch levels, the proposed framework remains relatively stable, indicating that the residual-guided representation does not require a perfectly accurate nominal reference in order to remain informative. As the mismatch grows, however, discrimination weakens, calibration quality deteriorates and nuisance alarms become more frequent. This behavior is consistent with the fact that residuals under model-plant mismatch contain both genuine fault information and spurious deviation introduced by reference error.

Table 12. Robustness under controlled reference-model mismatch. The nominal reference prediction is perturbed according to Eq. 31 before residual computation.

Mismatch level	F1	ECE	FAR/h	AUC	Interpretation
Clean reference, $\lambda = 0.00$	0.93	0.03	0.08	1.00	Baseline condition
Low mismatch, $\lambda = 0.05$	0.92	0.04	0.09	0.99	Mild degradation with stable operation
Moderate mismatch, $\lambda = 0.10$	0.90	0.05	0.11	0.97	Noticeable degradation but still usable
High mismatch, $\lambda = 0.20$	0.87	0.07	0.14	0.94	Strong degradation with increased threshold sensitivity

Figure 15 provides a visual summary of the same trend using side-by-side line plots versus mismatch magnitude. Panel (a) reports the discrimination-related metrics F1 and AUC, while panel (b) reports the calibration and false-alarm metrics ECE and FAR/h. Similar to the controlled perturbation results in Section 4.9, the framework remains tolerant to modest degradation, but not insensitive to it. This is an important practical finding because it indicates that the method can still operate when the nominal model is only approximate, while also confirming that calibration and threshold governance become increasingly important as mismatch grows.



(a) F1 and AUC versus mismatch magnitude. (b) ECE and FAR/h versus mismatch magnitude.

Figure 15. Performance under controlled reference-model mismatch. The horizontal axis shows the mismatch magnitude λ defined in Eq. 31. Panel (a) shows progressive degradation in F1 and AUC as mismatch increases, while panel (b) shows worsening calibration and increased nuisance alarms through ECE and FAR/h.

Taken together, these results show that the proposed residual-guided and calibrated detector does not depend on perfect model-plant agreement, but its performance degrades progressively as the nominal reference becomes less accurate. This supports the view that residuals should be treated as informative but imperfect signals whose operational usefulness depends on both reference quality and downstream decision governance.

- **Comparison with baseline models**

To assess the effectiveness of the proposed framework, we compare it against the baseline models introduced in Section 3.10. Table 13 reports discrimination, calibration and governance-oriented metrics for all methods. In addition to F1-score and AUC, we report ECE to assess probability quality, FAR/h to quantify nuisance alarms and median detection delay to reflect early-warning usefulness.

The proposed Physics-Guided CRNN provides the most balanced performance across discrimination, calibration and operational alarm control. Classical baselines remain competitive in simpler cases but have limited capacity to model longer temporal dependencies. Sequential deep baselines improve detection quality, but they still lack the explicit residual guidance, calibration layer or threshold-governance structure of the proposed framework.

Table 13. Comparison of baseline models and the proposed method under the same binary fault-detection protocol. Bold values indicate the best performance across models.

Model	F1-score	AUC	ECE	FAR/h	Median delay	Notes
Logistic Regression	0.84	0.90	0.11	0.15	60 steps	Linear baseline
Support Vector Machine	0.87	0.93	0.09	0.13	48 steps	Margin-based classifier
Random Forest	0.89	0.94	0.08	0.11	40 steps	Nonlinear tree ensemble
LightGBM	0.91	0.96	0.07	0.10	36 steps	Strong tabular baseline
LSTM	0.90	0.95	0.08	0.10	35 steps	Sequential baseline
CNN-LSTM	0.92	0.97	0.06	0.09	32 steps	Hybrid temporal baseline
Proposed Physics-Guided CRNN	**0.93**	**1.00**	**0.03**	**0.08**	**28 steps**	Residual-guided + calibrated

All models in Table 13 were trained under the same global binary labeling protocol rather than separately for each fault mode. Their reported scores therefore reflect evaluation under the same partition and decision policy, so the observed differences are attributable to the modeling approach rather than to different training regimes.

- **Summary of findings**

Across the evaluation suite, the proposed physics-guided CRNN shows stable training dynamics (Figure 7), strong latent separability (Figure 8) and near-ideal discrimination behavior (Figure 9). Calibration through Eq. 18 improves reliability (Figure 11) and strengthens probability quality under multiple metrics (Table 10), which in turn enables auditable threshold governance through Eq. 24 as illustrated in Figure 12. Per-fault results in Table 8 show strongest performance for supply, pressure and mechanical faults, while random variation remains the most challenging family. Operator-facing explanations in Figures 13 and 14 further support practical handoff by linking alarms to dominant sensors and residual descriptors and the robustness analysis in Table 11 shows graceful degradation under moderate noise and partial sensor loss.

• Discussion and comparison

This section interprets the reported results in relation to prior Tennessee Eastman Process studies, practical deployment requirements and the limitations of residual-centric, calibrated anomaly detection for industrial monitoring.

- **Comparison with literature**

To contextualize the proposed physics-guided convolutional recurrent neural network (CRNN), Table 14 summarizes represen-tative Tennessee Eastman Process studies spanning compact networks, attention-based convolutional neural network (CNN) variants and incipient-fault-focused designs. Reported performance varies with window length, preprocessing, class definition and whether the evaluation targets all faults or only weak incipient regimes such as IDV3, IDV9 and IDV15. These protocol differences reinforce the need to report deployment-relevant metrics and fault-wise behavior rather than relying on a single aggregate score.

Across recent deep baselines, compressed feedforward networks typically report accuracy in the mid-90% range, whereas recurrent and autoencoder-based hybrids often improve temporal modeling but remain sensitive to excitation assumptions and horizon design. Attention-based CNN pipelines that encode temporal windows into richer representations generally report stronger aggregate performance under fixed split protocols. This pattern is consistent with the latent separability in Figure 8 and the stable optimization behavior in Figure 7.

Under the split and end-to-end pipeline adopted in this work, the proposed CRNN achieves the strongest overall accuracy among the compared studies while avoiding image conversion and external excitation. More importantly, it augments strong discrimination with calibrated probabilities for auditable thresholding. The main contribution is therefore not only improved benchmark performance, but a more deployment-oriented combination of residual-guided learning and decision governance.

Table 14. Comparison of representative Tennessee Eastman Process reports. The reported metric is taken as stated in each source. When a study targets only incipient faults or reports a different primary metric, this is stated explicitly to avoid overstating comparability.

Method / Study	Year	Model family	Reported overall result
Temporal DL for TE fault detection[58] study	2021	Temporal deep model	Fault detection focus, metric per
Hierarchical Deep LSTM[59]	2022	Long short-term memory (LSTM) hierar-chical model	Acc. ≈93.4% on all faults
CNN with attention mechanism[60]	2023	CNN + attention	Acc. ≈98.46% on all faults
Dense feature ensemble net for incipient fault detection[61]	2024	Dense feature ensemble network	Incipient fault detection focus, metric per study
Granger-causal GAT for FD and RCD[62]	2024	Graph attention network	Fault detection and root-cause diagnosis focus, metric per study
CRNN (This work)	2026	Residual-guided CRNN + calibration	**Acc. = 99.0%** under this protocol

- **Overview**

The results show that combining physics-based residual evidence with a lightweight convolutional-recurrent backbone yields strong and stable anomaly detection on the Tennessee Eastman Process. Residual construction directs learning toward deviations from expected behavior while preserving channel-level interpretability and the Conv1D-bidirectional gated recurrent unit (BiGRU)-attention encoder captures both short- and long-range temporal structure under regime variability. As a result, the framework supports not only accurate detection but also probability outputs that are suitable for operational decision-making.

- **Implications for practice**

For practical deployment, the main lesson is that performance depends on the discipline of the full pipeline rather than on a single model block. Residual-enriched inputs improve separability and provide clearer operator-facing deviation signals, which is consistent with the gains reported in Table 7 and the family-wise trends in Table 8. Calibration then converts these scores into probabilities that support policy-driven threshold selection under explicit nuisance-alarm budgets, as shown in Figure 12. Together, these design choices support an alarm workflow that is both accurate and governable.

- **Limitations and threats to validity**

The evaluation is anchored to a single simulated benchmark and a binary detection objective, which limits external validity. Although the proposed pipeline combines residual generation, temporal modeling and calibration, as formalized in Eq. 8, Eq. 18

and Eq. 24, the reported results should be interpreted within the scope of the Tennessee Eastman benchmark rather than as direct proof of field-ready deployment. Real industrial plants may exhibit unmodeled disturbances, actuator nonlinearities, parameter drift, sensor degradation, operating-mode transitions and control interactions that are not fully captured in the present experiments.

An important practical limitation concerns the quality of the nominal reference model used for residual generation. As defined by Eq. 8, the residual is formed as the difference between the measured process output and the nominal reference-model prediction, but in real plants such a reference is rarely perfect. Model-plant mismatch, unmodeled dynamics, sensor bias, initialization error and external disturbances can all produce nonzero residuals even under healthy operation, thereby increasing nuisance alarms or weakening fault sensitivity. To examine this issue more directly, we introduced a controlled reference-model mismatch experiment in Section 4.10. The results in Table 12 and Figure 15 show that the proposed detector remains relatively stable under low mismatch levels, but discrimination, calibration quality and nuisance-alarm control degrade progressively as reference error increases. The proposed framework therefore does not assume that residuals are fault-exclusive signals; instead, it treats them as informative but imperfect inputs whose usefulness depends on the quality of the nominal reference together with downstream temporal modeling, post hoc calibration in Eq. 18 and operating-threshold selection in Eq. 24.

This point is important when interpreting the governance and calibration results in Section 4.7 and Section 4.8. The threshold sweep in Figure 12 and the operating-point summary in Table 9 show that explicit threshold governance is necessary because alarms are influenced by nuisance deviations as well as genuine fault signatures. Likewise, the calibration improvements reported in Table 10 should be interpreted as improving probability quality under imperfect residual conditions, not as evidence of perfect model-plant agreement.

Finally, calibration is performed offline and may drift during operation, especially under regime shifts, sensor aging or maintenance-related changes in process behavior. The explanation results in Figure 13 and Figure 14 improve interpretability, but they do not remove the underlying dependence of the framework on the quality of the nominal reference model and the representativeness of the benchmark data. These limitations motivate future work on adaptive recalibration, drift monitoring and evaluation under stronger plant-model mismatch conditions.

- **Operational guidance**

In deployment, thresholds should be selected from calibrated probabilities using an explicit nuisance-alarm budget and verified through threshold-sensitivity analysis. A practical default is to operate at the selected $\delta^\star$, monitor early-warning delay and nuisance alarms per hour as routine health indicators and update thresholds or recalibrate when drift is detected. A conservative rollout strategy is to begin in shadow mode and promote the system to action-triggering workflows only after stability has been confirmed under noise and partial sensor loss.

Table 15 summarizes the main method-to-evidence-to-next-step mapping. Residual evidence strengthens sensitivity and interpretability, calibrated threshold governance makes alarm policies reproducible and explanation summaries support rapid operator triage. The most important next steps are extending beyond the benchmark setting, strengthening drift monitoring and evaluating robustness under broader shift conditions.

Table 15. Structured discussion summary aligned with the reported results and the proposed pipeline.

Aspect / Theme	Type	Evidence in this study	Next step / extension
Residual evidence and modeling			
Residual enrichment	Physics-informed	Improves detection versus the base model and supports clearer fault-family separation.	Track residual drift and refit the reference model and scalers under controlled procedures.
Temporal encoder	Deep temporal	Captures short- and long-range temporal structure and yields stable optimization behavior.	Stress-test window length and add mode-aware validation where possible.
Calibration, governance and explanations			
Probability calibration	Governance layer	Improves reliability and enables stable, policy-driven operat-ing thresholds.	Monitor calibration drift and schedule periodic recalibration.
Threshold policy	Risk control	Threshold sweep exposes FAR-TPR trade-offs and supports auditable $\delta^\star$ selection.	
Operator explanations	Interpretability	Top-feature and per-fault snapshots identify dominant chan-nels for investigation.	

Use mode- or family-specific alarm budgets if required.		
Test attribution stability and group features by subsystem for triage.		
Deployment and external validity		
Robustness	Shift tolerance	Controlled perturbations show graceful degradation under moderate noise and dropout.
Add drift detection and sensor-health monitoring in produc-tion.		
Scope	Benchmark limit	
Single simulated benchmark and binary objective limit exter-nal validity.		
Validate on additional datasets and real plant logs where avail-able.		

• Conclusion and Future Work

This work addressed key deployment barriers in industrial anomaly detection by developing and validating a calibrated, physics-guided pipeline on the Tennessee Eastman Process benchmark. The framework integrates twin-based residual construction with compact rolling and spectral descriptors and then models the resulting residual-enriched windows using an attention-based convolutional recurrent neural network (CRNN). SHapley Additive exPlanations (SHAP)-guided feature selection reduces redundancy and improves sensor-level interpretability, while Optuna-based tuning stabilizes training and yields a compact configuration suitable for practical monitoring. Under the run-level evaluation protocol, the method delivers strong discrimination and consistent fault-family behavior, with the largest gains observed for supply, pressure and mechanical regimes, while random-variation faults remain the most challenging.

Beyond discrimination, the study emphasizes decision governance and audit readiness. Platt scaling improves probability reliability, enabling explicit threshold selection under a nuisance-alarm budget and supporting early-warning operation with controlled false alarms. Statistical validation via block bootstrap confidence intervals provides uncertainty-aware evidence for the reported performance and explanation outputs from SHAP and Integrated Gradients connect alarms to dominant sensors for operator triage. Overall, the results show that residual-guided learning combined with calibrated probabilities can transform a high-performing detector into a policy-driven monitoring tool that is easier to justify, tune and hand off in Industry 4.0 environments.

• Future Work

Future research can extend this work in several directions. Physics-based models can be expanded to incorporate multiple data sources, such as sound, vibration, images and control logs, so that complementary evidence can improve detection under more complex fault conditions. Federated learning and continual learning are also promising directions for cross-plant deployment and long-term adaptation. Federated learning enables model development without sharing raw plant data, while continual learning can help the system adapt to process drift, wear and evolving operating conditions.

Another important direction is the design of lightweight CRNN variants for on-device deployment. Model-compression techniques such as pruning, quantization and knowledge distillation can make real-time detection feasible on low-power industrial edge devices. Together, these directions can support the development of a reliable, efficient and more generalizable CRNN-based monitoring system for modern smart-factory environments. Future work will also extend the present binary anomaly detector toward multiclass fault diagnosis so that the framework can not only detect abnormal operation but also identify the specific fault type.

Nomenclature

Variables and Symbols

Table 16. Variables and symbols used in the proposed framework.

Acronyms and Abbreviations

Author Contributions

Abuzar Khan led the conceptualization of the study, designed the methodology, conducted the core experiments and analysis and wrote the original draft of the manuscript. Fahmid Al Farid contributed to data analysis, experimental support and manuscript review. Ahmad Junaid contributed to experimental execution, data analysis and software implementation. Abid Iqbal supported software development, data curation and experimental validation. Muhammad Farooq Siddique and Muhammad Shahzad

Table 17. Acronyms and abbreviations used in the manuscript.

Acronym	Meaning	Acronym	Meaning	Acronym	Meaning
TEP	Tennessee Eastman Process	CRNN	Convolutional Recurrent Neural Net-work	CNN	Convolutional Neural Network
Conv1D	One-dimensional convolution	BiGRU	Bidirectional gated recurrent unit	GRU	Gated recurrent unit
LSTM	Long short-term memory	SHAP	SHapley Additive exPlanations	LightGBM	Light Gradient Boosting Machine
ECE	Expected calibration error	ROC	Receiver operating characteristic	AUC	Area under the curve
AUC-ROC	Area under ROC curve	PR	Precision-recall	AUC-PR	Area under PR curve
F1	F1-score	NAB	Numenta Anomaly Benchmark	BCa	Bias-corrected and accelerated
FAR/h	False alarm rate per hour	RBF	Radial basis function	SVM	Support vector machine
RF	Random forest	LR	Logistic regression	IDV	Input disturbance variable
IoT	Internet of Things	NLL	Negative log-likelihood	MCE	Maximum calibration error

Siddique contributed to validation, technical verification and critical review of the results. Jia Uddin provided additional support in data analysis, result interpretation and manuscript refinement. Hezerul Abdul Karim contributed to funding acquisition, methodological guidance, validation and critical review of the manuscript. Ghassan Husnain supervised the study, coordinated project management and led the critical review and editing process. All authors reviewed and approved the final manuscript.

Conflict of Interest

The author declares no conflict of interest related to the publication of this work.

Data Availability

The Tennessee Eastman Process (TEP) dataset used in this study is publicly available and can be accessed through established industrial benchmark repositories or via the references cited in the manuscript. It is also available from Kaggle: Dataset HYPERLINK "https://www.kaggle.com/datasets/averkij/tennessee-eastman-process-simulation-dataset"__ HYPERLINK "https://www.kaggle.com/datasets/averkij/tennessee-eastman-process-simulation-dataset"Link.

Code Availability

The code used to reproduce the experiments, result-generation pipeline and analysis presented in this study is publicly available at GitHub HYPERLINK "https://github.com/abuzarkhaaan/TEP_Physcis_CRNN" HYPERLINK "https://github.com/abuzarkhaaan/TEP_Physcis_CRNN"repository. The repository includes the main notebook, supporting result folders and implementation files required for the proposed Tennessee Eastman Process fault-detection framework.

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Ethics statement

This study uses the publicly available *Tennessee Eastman Process (TEP)* dataset[55,56], which is a well-established industrial benchmark for process monitoring and fault diagnosis. The data are fully simulated and do not involve any human participants, personal data or animal experiments. Therefore, no ethical approval was required.

Ethical Approval

Not applicable.

Informed Consent Statement

Not applicable.

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